D3A-Net: A Deformable 3-D Attention Network for Hyperspectral Unmixing

Fanqiang Kong, Haocheng Yang, Zhibo Wu, Tongshu Wu, *Student Member, IEEE*, and Yunsong Li, *Member, IEEE*

***Abstract*—Hyperspectral unmixing (HU) aims to decompose mixed pixels into endmember spectra and corresponding abundance maps under physically interpretable constraints. Despite recent progress in convolutional neural network (CNN)- and Transformer-based approaches, most existing methods still rely on rigid sampling grids or process spatial and spectral features separately, which may cause feature misalignment for irregular and elongated structures and weaken spectral–spatial volumetric continuity. To address these limitations, we propose a deformable 3-D attention network (D3A-Net), a fully 3-D U-shaped encoder–decoder that processes hyperspectral imagery as a unified data cube and overcomes the constraints of fixed receptive fields. D3A-Net is constructed by stacking 3D-DeFormer blocks (3D-DFBs), each of which incorporates a 3-D multiscale deformable neighbourhood-attention (MS-DNA) module and a complementary local extraction branch, followed by a gated feed-forward module for efficient feature transformation. The physics-guided unmixing Encoder II enforces the abundance nonnegativity and sum-to-one constraints, while the linear decoder reconstructs the hyperspectral image (HSI), enabling end-to-end unsupervised learning. Extensive experiments on a synthetic dataset under multiple signal-to-noise ratio (SNR) settings and three real-world datasets (Samson, Jasper Ridge, and APEX) demonstrate that D3A-Net achieves accurate endmember estimation and abundance estimation.**

***Index Terms*—3-D encoder–decoder, deformable attention, hyperspectral unmixing (HU), spectral–spatial learning, unsupervised learning.**

I. Introduction

A HYPERSPECTRAL image (HSI) captures spectral information across dozens to hundreds of bands, spanning the visible to the infrared spectrum. By uniquely integrating spatial and spectral information, a hyperspectral image (HSI) has revolutionized Earth observation [1], [2]. This rich spectral resolution enables fine-grained material identification based on unique spectral signatures, thereby propelling the development of frontier applications such as precision agriculture [3], mineral mapping [4], environmental monitoring [5], and urban land-cover classification [6]. However, due to physical and technological constraints, the high spectral resolution of HSI is often achieved at the expense of spatial resolution [7]. This low spatial resolution, combined with the complexity of ground materials and atmospheric mixing effects, results in pixels that contain spectral information from multiple distinct materials. These are referred to as mixed pixels [8]. To enable effective quantitative analysis of HSI, hyperspectral unmixing (HU) has been extensively studied. The core objective of HU is to decompose mixed pixels into their constituent material spectra (endmembers) and estimate the proportion of each material (abundance) [9], [10].

Due to its physical interpretability and mathematical tractability, the linear mixing model (LMM) is widely adopted in most unmixing algorithms [11]. The LMM assumes that incident light interacts with ground materials only once before being captured by the sensor. Consequently, the observed mixed spectrum is treated as a linear superposition of the spectra of distinct materials. Based on this, a mixed pixel can be modeled as a linear combination of endmembers, subject to the abundance nonnegativity constraint (ANC) and the abundance sum-to-one constraint (ASC) [12].

Early unmixing research was dominated by geometric and statistical methods. Representative geometric algorithms include N-FINDR, an algorithm for fast autonomous spectral endmember determination [13], and vertex component analysis (VCA) [14]. These methods assume the presence of pure pixels in the data and seek to identify the vertices of the corresponding data simplex. Despite their high computational efficiency, the performance of these methods degrades rapidly in highly mixed scenarios where pure pixels are absent [15]. To mitigate this issue, statistical methods such as independent component analysis (ICA) [16] and nonnegative matrix factorization (NMF) [17] were introduced. Furthermore, sparse-regression methods based on spectral libraries, such as sparse unmixing via variable splitting and augmented Lagrangian (SUnSAL) [18], [19], have also been widely employed. Although these traditional model-based methods are theoretically elegant, they typically treat HSI as a collection of unstructured, isolated spectral vectors, treating each pixel spectrum independently [20]. This approach inherently discards the rich spatial correlations and texture information embedded in the image domain, thereby limiting the robustness of these algorithms in the presence of noise and outliers [21].

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The rise of deep learning (DL) has fundamentally reshaped the field of HU, providing powerful tools for modeling nonlinear spectral distortions and learning robust feature representations [22]. Autoencoder (AE)-based networks, which map spectra to a latent abundance space (encoder) and reconstruct them back to the original space (decoder), have become the dominant paradigm for unsupervised unmixing [23]. Early 1D-convolutional neural network (CNN) methods primarily focused on spectral feature extraction [24], [25]. Recognizing the crucial role of spatial context, subsequent researchers introduced 2-D and 3-D CNNs to exploit spatial–spectral dependencies. For instance, the convolutional AE for spectral–spatial hyperspectral unmixing (CNNAEU) [26] utilizes a convolutional encoder to aggregate neighborhood spectral–spatial information. Subsequently, more sophisticated architectures addressing specific unmixing challenges have emerged. CyCU-Net [27] introduces cycle-consistency constraints via cascaded AEs to optimize endmember discrimination, MSNet [28] adopts a multistage progressive learning strategy to capture features from coarse to fine resolutions, and SSAF-Net [29] integrates a dual-branch encoder to explicitly model endmember variability (EV). Meanwhile, DFFN improves spatial information utilization and reduces initialization sensitivity through dual-feature fusion and direct endmember estimation [30], whereas A2SAN employs abundance-guided spatial and spectral attention to enhance material-relevant representations [31]. Recent dual-branch methods include DASS-Net, which combines parallel spatial and spectral streams using neighborhood spatial attention and dual-gated fusion [32], and MCDB-Net, which integrates full-view and local multiview spectral features through a multiview abundance collaboration module [33]. These methods highlight the value of complementary spectral–spatial representations; however, their feature interactions are organized through parallel streams or predefined spectral views, rather than learned coordinate sampling in the native 3-D spectral–spatial volume. Consequently, fixed-grid feature extraction remains a common limitation when modeling irregular material geometry or spatially disjoint yet spectrally similar regions [34].

To overcome the locality bias of convolutional architectures, Transformer-based models have recently been introduced into HU [35] to capture long-range dependencies and nonlocal material correlations. DeepTrans integrates a Transformer into an AE framework to enhance latent feature representation for the joint estimation of abundances and endmembers [36]. More recently, the self-supervised relation-aware Vision Transformer (SRViT) employs a self-embedded module, pixel-level positional encoding, and self-supervised contrastive learning to preserve local spatial correlations across different views and alleviate the spatial discontinuity caused by conventional nonoverlapping patch partitioning [37]. The sampling-based spatial–spectral transformer and generative adversarial network (SSST-GAN) adopts dual-branch sampling-based Transformer encoders, in which full spatial sampling and sparse spectral sampling are, respectively, used to extract complementary spatial and spectral representations [38]. It further combines feature enhancement, nonlinear decoding, and adversarial learning to improve reconstruction fidelity and unmixing accuracy.

These studies demonstrate the strong potential of Transformer architectures for global dependency modeling and spatial–spectral feature learning. Nevertheless, many existing methods still organize feature interactions according to fixed token layouts or predefined spatial and spectral sampling strategies. Their sampling coordinates are therefore not explicitly adapted to irregular, elongated, or anisotropic material distributions. Moreover, spatial and spectral representations are frequently extracted through separate branches, which may not fully preserve the native volumetric continuity of hyperspectral data. These limitations motivate the development of a unified 3-D deformable-attention framework that learns continuous spectral–spatial sampling offsets according to the underlying scene content.

Standard CNNs and Vision Transformers operate on fixed, regular sampling grids. CNNs slide fixed square convolutional kernels, while Transformers partition images into fixed rectangular patches [39]. This geometric rigidity causes severe misalignment when processing features with irregular, elongated, or anisotropic shapes—such as winding roads, rivers, or scattered vegetation boundaries—which are prevalent in datasets like Jasper Ridge and APEX. As noted in recent computer vision research [40], [41], when a fixed square window processes a slender target, it inevitably samples a majority of pixels from the surrounding background, such as soil and water. This effect dilutes the target's feature representation, causing the unmixed abundance maps of fine-grained objects to appear fractured, blurred, or spectrally distorted. Existing methods lack the flexibility to spatially deform their sampling positions to conform to these complex boundaries.

Additionally, hyperspectral data are intrinsically a 3-D data cube composed of two spatial dimensions and one spectral dimension. Ideally, feature extraction should be performed within this native 3-D space to preserve local spectral–spatial consistency [42], [43]. However, to reduce computational complexity, many existing networks adopt "pseudo-3D" strategies, such as separating spatial 2D-CNNs and spectral 1D-CNNs, or flattening the spectral dimension into channels. While computationally efficient, these methods risk severing the coupled correlations between spectral bands and spatial textures [44]. Although true 3D-CNN-based HU models have been developed, representative methods commonly employ stacked dense 3-D convolutions over highly overlapping spectral–spatial patches to jointly capture spatial and spectral information [45]. Such patch-wise processing repeatedly evaluates neighboring voxels and therefore incurs substantial arithmetic and memory costs. Moreover, their fixed convolutional kernels remain geometrically rigid when modeling irregular or elongated material distributions.

To address these geometric and dimensional challenges, we propose a deformable 3-D attention network (D3A-Net). D3A-Net is a fully 3-D architecture designed to respect the volumetric nature of HSI while breaking the constraints of fixed-sampling grids. Inspired by deformable mechanisms [46], [47], we introduce a 3-D multiscale

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deformable neighbourhood-attention (MS-DNA) module tailored for unmixing. Unlike standard attention mechanisms that scan fixed patches, MS-DNA learns to generate continuous 3-D coordinate offsets, dynamically shifting its attention sampling points to cover the effective regions of objects while filtering out background noise. To ensure that this flexibility does not come at the cost of high-frequency detail loss, we design a parallel dual-branch module 3D-DeFormer block (3D-DFB). This module combines the global, shape-adaptive modeling capability of deformable attention with the local spectral–spatial detail preservation capability of a complementary 3-D local extract branch.

Accordingly, D3A-Net differs from existing Transformer-based HU methods by constructing the key and value tensors from features sampled at query-conditioned 3-D coordinates, rather than relying only on fixed token layouts or predefined spatial–spectral sampling patterns. Compared with 3D-CNN-based HU, MS-DNA preserves volumetric spectral–spatial modeling while allowing the effective sampling support and aggregation weights to vary with the input content. In summary, the main contributions of this article are as follows.

1) Efficient Dual-Branch 3-D Framework: We propose a novel network termed D3A-Net, which features a global–local dual-branch architecture. This design effectively extracts long-range global context while simultaneously preserving fine-grained local details and textures in HSI.

2) Spatial–Spectral Learning via 3-D Neighborhood Attention: By leveraging 3-D neighborhood attention with linear complexity, the proposed framework processes HSI as a unified volumetric data cube and captures intrinsic spectral–spatial correlations. Compared with dense 3D-CNNs using fixed cubic kernels, the bounded-neighborhood attention and lightweight pointwise/depthwise 3-D operations reduce repeated patch-wise computation while preserving volumetric feature interactions.

3) Shape-Adaptive Unmixing via MS-DNA: We introduce the MS-DNA module to HU. It predicts continuous offsets along the spectral and two spatial axes and constructs the key and value tensors from features sampled at query-conditioned coordinates. The resulting content-dependent receptive field improves feature alignment for irregular or slender material distributions.

The remainder of this article is organized as follows. Section II introduces the problem formulation and the LMM. Section III elaborates on the proposed D3A-Net, detailing the 3-D-DeFormer blocks and the MS-DNA mechanism. Section IV presents extensive experimental results on both synthetic and real-world datasets, along with a comprehensive performance analysis. Finally, Section V concludes this study.

II. Problem Formulation

Let $\mathcal{Y} \in \mathbb{R}^{L\times H\times W}$ denote an HSI cube with L spectral bands and spatial size $H \times W$. After reshaping into a matrix, we have $\mathbf{Y} \in \mathbb{R}^{L\times N}$, where $N = H \times W$ is the number of pixels, and the ith column $\mathbf{y}_i \in \mathbb{R}^L$ is the observed spectrum of pixel i. Under the LMM, each pixel is modeled as

$$\mathbf{y}_i = \sum_{j=1}^{P} a_{j,i}\mathbf{m}_j + \mathbf{n}_i \tag{1}$$

where P is the number of endmembers, $\mathbf{m}_j \in \mathbb{R}^L$ is the jth endmember spectrum, $a_{j,i}$ is its abundance at pixel i, and $\mathbf{n}_i$ is noise.

By aggregating all endmembers into a matrix $\mathbf{M} = [\mathbf{m}_1, \ldots, \mathbf{m}_P] \in \mathbb{R}^{L\times P}$ and all abundances into a matrix $\mathbf{A} = [\mathbf{a}_1, \ldots, \mathbf{a}_N] \in \mathbb{R}^{P\times N}$, the model becomes

$$\mathbf{Y} = \mathbf{MA} + \mathbf{N}. \tag{2}$$

To satisfy physical interpretability, abundances are constrained by ANC and ASC

$$a_{j,i} \geq 0, \quad \sum_{j=1}^{P} a_{j,i} = 1. \tag{3}$$

III. Method

In this section, we elaborate on the proposed D3A-Net. First, we present the overall architecture designed for volumetric hyperspectral data. Subsequently, we describe the core 3D-DFB, which integrates a 3-D MS-DNA module, a complementary local extraction branch, and a SiLU–GRN gated feed-forward network (SGG-FFN), where SiLU and GRN denote the sigmoid linear unit and global response normalization, respectively. Finally, we introduce the physics-guided unmixing encoder, the linear reconstruction decoder, and the loss function of our method.

A. Network Overview

Unlike traditional methods that flatten spectral bands or process spatial–spectral dimensions sequentially, D3A-Net treats the HSI as a volumetric 3-D data cube to preserve intrinsic spectral–spatial correlations. As illustrated in Fig. 1, the network adopts a U-shaped hierarchical architecture structured into three functional stages: Encoder I, Encoder II, and Decoder. First, the input HSI batch $\mathcal{Y}_{\text{in}} \in \mathbb{R}^{B\times L\times H\times W}$ is reshaped into a 5-D volumetric tensor $\mathcal{Y}_{\text{vol}} \in \mathbb{R}^{B\times 1\times L\times H\times W}$ by interpreting the spectral dimension as the depth axis. Then, a $1 \times 1 \times 1$ 3-D convolution projects $\mathcal{Y}_{\text{vol}}$ into the initial feature tensor $\mathcal{X}_0 \in \mathbb{R}^{B\times C\times D\times H\times W}$, where $D = L$. The core of D3A-Net consists of stacked 3D-DFBs. Encoder I progressively reduces the spatial resolution while increasing channel capacity using a downsampling module. This module decouples channel projection from spatial scaling by first projecting features via a $1 \times 1 \times 1$ convolution with Gaussian error linear unit (GELU) activation, followed by a $3 \times 3 \times 3$ convolution with a stride of $(1, 2, 2)$ for spatial downsampling, which effectively preserves spectral integrity. The subsequent upsampling path restores resolution via a feature-refinement upsampling module. By integrating trilinear interpolation with anti-aliasing depthwise convolutions, root-mean-square layer normalization (RMSNorm), and GELU, this module suppresses checkerboard artifacts while preserving fine-grained details. Symmetric skip connections fuse low-level details from Encoder I with high-level semantic features

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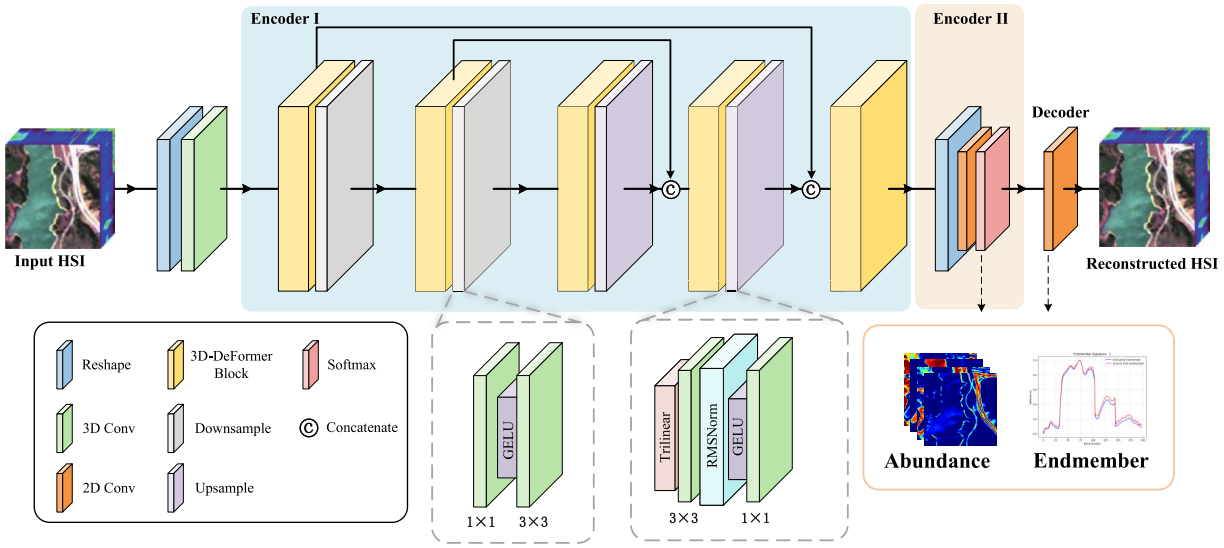


Fig. 1. Overall architecture of D3A-Net. Encoder I extracts hierarchical 3-D spectral–spatial features using stacked 3D-DFBs and spatial downsampling/upsampling operations with skip connections. Encoder II maps the fused features to abundance maps satisfying the ANC and ASC, and the linear decoder reconstructs the input HSI, with its weights representing the estimated endmember spectra.

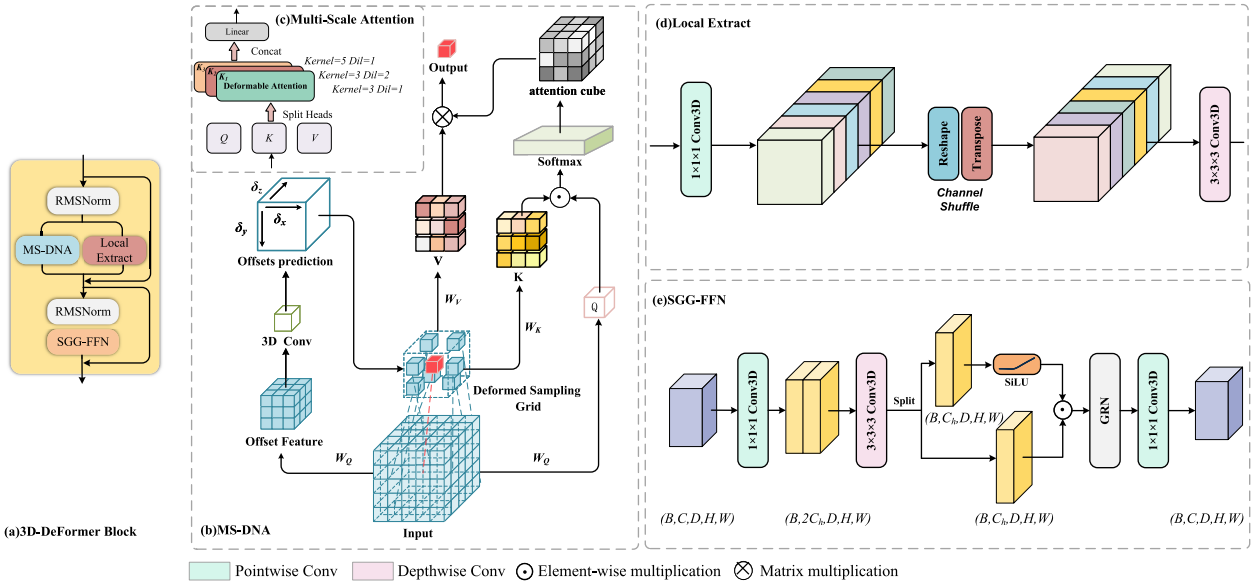


Fig. 2. Detailed architecture of the 3-D-DeFormer block and its components. (a) 3D-DFB employs parallel MS-DNA and local extraction branches, followed by the SGG-FFN with residual connections. (b) MS-DNA predicts 3-D offsets from the query tensor, samples features at the deformed coordinates to construct the keys and values, and performs neighborhood-attention aggregation. The red cube inside the deformed grid denotes a displaced sampling position, whereas the red cube labeled "Output" denotes the attention-aggregated output feature. (c) Multiscale attention module divides the attention heads into three groups with different kernel-size and dilation configurations, and then concatenates and projects their outputs. (d) Local extraction branch sequentially applies pointwise convolution, channel shuffle, and depthwise 3-D convolution to preserve local spectral–spatial details. (e) SGG-FFN expands the feature channels, applies depthwise convolution, SiLU-based gating, and GRN, and finally projects the features back to the original channel dimension.

to strengthen multiscale representation learning. Encoder II then maps the recovered deep features to abundance maps under physical constraints. Finally, the Decoder reconstructs the HSI from the estimated abundances, yielding physically interpretable unmixing outputs.

B. 3D-DeFormer Block

The core unit of D3A-Net is the 3D-DFB. As shown in Fig. 2(a), the block consists of two parallel branches, namely, the global MS-DNA branch and the complementary local extraction branch, followed by the SGG-FFN. The remaining panels provide module-level expansions of this block-level flow rather than independent processing stages. Fig. 2(b) details MS-DNA; Fig. 2(c) further expands the multiscale head grouping within MS-DNA; and Fig. 2(d) and (e) details the complementary local extract and SGG-FFN, respectively. As formalized in (4), the outputs of the two parallel branches in Fig. 2(b) and (d) are added to the residual input; the resulting feature is then processed by the SGG-FFN in Fig. 2(e) with

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a second residual connection, as given in (5). Let $\mathcal{X}_{l-1} \in \mathbb{R}^{B\times C\times D\times H\times W}$ denote the input tensor of the lth block. The interaction process is formulated as

$$\widetilde{\mathcal{X}}_l = \text{MS-DNA}\left(\text{RMSNorm}\left(\mathcal{X}_{l-1}\right)\right) + \text{Local}\left(\text{RMSNorm}\left(\mathcal{X}_{l-1}\right)\right) + \mathcal{X}_{l-1} \tag{4}$$

$$\mathcal{X}_l = \text{SGG-FFN}\left(\text{RMSNorm}\left(\widetilde{\mathcal{X}}_l\right)\right) + \widetilde{\mathcal{X}}_l \tag{5}$$

where l denotes the index of the 3D-DFB and RMSNorm$(\cdot)$ denotes rms normalization adapted to 3-D feature tensors. For clarity, the normalized intermediate tensor is denoted by

$$\mathcal{Z}_{\text{in},l} = \text{RMSNorm}\left(\widetilde{\mathcal{X}}_l\right) \tag{6}$$

which serves as the input of the subsequent SGG-FFN.

1) 3-D Deformable Neighborhood Attention: Standard CNNs operate on fixed geometric grids, limiting their efficacy in modeling the irregular and anisotropic distributions of ground materials in HSIs. While neighborhood attention reduces the quadratic complexity of global self-attention to linear complexity, its receptive field remains confined to a rigid sliding window. To address these geometric limitations, we propose the MS-DNA mechanism, as illustrated in Fig. 2(b). This module dynamically learns to deform the sampling grid, aligning it with the semantic content of the input features. Let $\mathcal{X} \in \mathbb{R}^{B\times C\times D\times H\times W}$ denote the input feature tensor, and the query feature is obtained as $\mathcal{Q} = \mathcal{X}\mathbf{W}_Q$, where $\mathcal{Q} \in \mathbb{R}^{B\times C\times D\times H\times W}$. Given the query feature $\mathcal{Q}$, the dense 3-D offset field is predicted as

$$\Delta\mathcal{P} = \tanh\left(\Phi_{\text{offset}}\left(\mathcal{Q}\right)\right) \cdot s \tag{7}$$

where $\Delta\mathcal{P} \in \mathbb{R}^{B\times 3\times D\times H\times W}$ denotes the dense 3-D offset tensor, $\Phi_{\text{offset}}(\cdot)$ denotes the lightweight 3-D offset-prediction subnetwork, and s is a learnable scaling scalar controlling the magnitude of the predicted offsets. The three channels of $\Delta\mathcal{P}$ correspond to the spectral, vertical spatial, and horizontal spatial displacements, denoted by δ_z, δ_y, and δ_x in Fig. 2(b), respectively. The tanh function constrains the predicted offsets to the range $(-1, 1)$, while s adjusts their overall magnitude, thereby stabilizing the training process and preventing sampling divergence.

For a regular reference coordinate $\mathbf{p}_{\text{reg}} \in \mathbb{R}^3$, the corresponding displacement vector extracted from $\Delta\mathcal{P}$ is denoted by $\Delta\mathbf{p} \in \mathbb{R}^3$. The deformed sampling coordinates are calculated as

$$\mathbf{p}_{\text{sample}} = \mathbf{p}_{\text{reg}} + \Delta\mathbf{p} \tag{8}$$

where $\mathbf{p}_{\text{reg}}$ denotes a coordinate on the regular reference grid and $\mathbf{p}_{\text{sample}}$ denotes the corresponding deformed sampling coordinate obtained after adding the learned offset. In Fig. 2(b), the red dashed line illustrates the displacement from the central position of the regular reference grid to the red cube within the light-blue deformed sampling grid. Therefore, this red cube represents a representative displaced sampling center $\mathbf{p}_{\text{sample}}$, rather than an output feature, while the surrounding light-blue cubes denote the corresponding 3-D deformed sampling neighborhood.

Since the learned sampling coordinates are generally fractional, trilinear interpolation is employed to obtain their feature values

$$\widetilde{\mathbf{x}}(\mathbf{p}) = \sum_{\mathbf{q}\in\mathcal{N}(\mathbf{p})} G\left(\mathbf{q}, \mathbf{p}\right)\, \mathbf{x}\left(\mathbf{q}\right) \tag{9}$$

where $\mathcal{N}(\mathbf{p})$ denotes the set of eight neighboring integer-grid coordinates of $\mathbf{p}$, $\mathbf{x}(\mathbf{q}) \in \mathbb{R}^C$ denotes the feature vector at the integer coordinate $\mathbf{q}$, $\widetilde{\mathbf{x}}(\mathbf{p}) \in \mathbb{R}^C$ denotes the interpolated feature vector at $\mathbf{p}$, and $G(\mathbf{q}, \mathbf{p})$ is the trilinear interpolation weight determined by the relative coordinate distances between $\mathbf{p}$ and $\mathbf{q}$ along the spectral, vertical spatial, and horizontal spatial dimensions. Stacking the interpolated feature vectors over all deformed sampling coordinates yields the resampled feature tensor $\mathcal{X}_{\text{samp}} \in \mathbb{R}^{B\times C\times D\times H\times W}$. Then, the deformed key and value tensors $\mathcal{K}, \mathcal{V} \in \mathbb{R}^{B\times C\times D\times H\times W}$ are, respectively, computed as

$$\mathcal{K} = \mathcal{X}_{\text{samp}}\mathbf{W}_K$$
$$\mathcal{V} = \mathcal{X}_{\text{samp}}\mathbf{W}_V \tag{10}$$

where $\mathbf{W}_K$ and $\mathbf{W}_V$ denote the corresponding learnable projection matrices. $\mathcal{Q}$ is then matched with $\mathcal{K}$ to compute the attention scores. Here, $\mathcal{Q}$ is computed on the regular volumetric lattice, whereas $\mathcal{K}$ and $\mathcal{V}$ are generated from $\mathcal{X}_{\text{samp}}$ after deformable resampling. Therefore, the effective neighborhood of each query is determined jointly by the learned 3-D offsets and the subsequent attention weights, rather than remaining fixed as in regular neighborhood attention or 3-D convolution. After Softmax normalization, these scores are used to perform weighted aggregation over $\mathcal{V}$. In Fig. 2(b), the gray attention cube visualizes the distribution of attention weights over the deformed 3-D neighborhood, whereas the red cube labeled "Output" denotes the output feature obtained after attention-weighted aggregation. Therefore, the red cube inside the deformed sampling grid represents a displaced sampling coordinate, while the red cube at the output side represents the aggregated output feature. The detailed multiscale grouping strategy and neighborhood configurations are introduced in Section III-B2.

2) Multiscale Attention Calculation: To simultaneously capture fine-grained spectral variations and long-range spatial dependencies within a unified attention module, we extend the basic deformable attention to a three-scale variant. The key insight is to leverage different receptive fields across different feature subspaces. We partition the total number of attention heads N_h into three parallel groups $\{G_1, G_2, G_3\}$. The three groups are associated with different receptive-field configurations

$$(k_1, r_1) = (k, r)$$
$$(k_2, r_2) = (k, 2r)$$
$$(k_3, r_3) = (k + 2, r) \tag{11}$$

where k denotes the spatial kernel size and r denotes the dilation rate. This architecture permits the network to perceive the input context at varying scales. These scales range from dense local details to sparse global contexts, preserving linear complexity with a small constant-factor overhead. Accordingly, for scale $s \in \{1, 2, 3\}$, $\mathcal{Q}_s$, $\mathcal{K}_s$, and $\mathcal{V}_s$ denote the

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corresponding head-group query, key, and value tensors. At voxel coordinates $\mathbf{p}$ and $\mathbf{q}$, the corresponding feature vectors are denoted by $\mathbf{q}_s(\mathbf{p})$, $\mathbf{k}_s(\mathbf{q})$, and $\mathbf{v}_s(\mathbf{q})$, respectively. The normalized attention weight is computed as

$$\alpha_s(\mathbf{p},\mathbf{q}) = \underset{\mathbf{q}\in\mathcal{N}_s(\mathbf{p})}{\text{Softmax}}\left(\frac{\mathbf{q}_s(\mathbf{p})^\top \mathbf{k}_s(\mathbf{q})}{\sqrt{d_h}} + b_{\text{rel}}^{(s)}(\mathbf{p},\mathbf{q})\right) \tag{12}$$

where $\mathcal{N}_s(\mathbf{p})$ denotes the neighborhood of $\mathbf{p}$ determined by (k_s, r_s), $d_h = C/N_h$ denotes the channel dimension of each attention head, and $b_{\text{rel}}^{(s)}(\mathbf{p},\mathbf{q})$ denotes the relative positional bias between $\mathbf{p}$ and $\mathbf{q}$ at scale s. The attention output vector at $\mathbf{p}$ is then obtained as

$$\mathbf{a}_s(\mathbf{p}) = \sum_{\mathbf{q}\in\mathcal{N}_s(\mathbf{p})} \alpha_s(\mathbf{p},\mathbf{q})\ \mathbf{v}_s(\mathbf{q}). \tag{13}$$

Stacking $\mathbf{a}_s(\mathbf{p})$ over all voxel coordinates produces the scale-specific output tensor $\mathcal{A}_s$. To preserve local positional information, a local positional encoding tensor is generated from the query tensor

$$\mathcal{E}_{\text{pos}} = \text{DWConv}_{3\times3\times3}(\mathcal{Q}) \tag{14}$$

where $\mathcal{E}_{\text{pos}} \in \mathbb{R}^{B\times C\times D\times H\times W}$ and $\text{DWConv}_{3\times3\times3}$ denotes a $3\times3\times3$ depthwise convolution. Finally, the outputs of the three-scale groups are concatenated and projected to fuse the multiscale spectral–spatial features

$$\mathcal{X}_{\text{out}} = \text{Concat}(\mathcal{A}_1,\mathcal{A}_2,\mathcal{A}_3)\,\mathbf{W}_o + \mathcal{E}_{\text{pos}} \tag{15}$$

where $\mathbf{W}_o$ denotes the learnable output projection matrix and $\mathcal{X}_{\text{out}} \in \mathbb{R}^{B\times C\times D\times H\times W}$ denotes the output tensor of the MS-DNA module. Although both neighborhood attention and 3-D convolution scale linearly with the number of voxels, MS-DNA aggregates features within a bounded neighborhood and combines pointwise projections with depthwise 3-D operations. This avoids the large channel-to-channel kernel cost and repeated overlapping-patch computation commonly incurred by dense 3D-CNN-based HU models.

3) Complementary Local Extract: While the MS-DNA module effectively addresses long-range deformations, preserving local spectral–spatial continuity remains imperative for accurate unmixing. To complement this, we design a local extraction branch. Fig. 2(d) shows the detailed structure of local extract. This module operates through a sequential process. First, we employ a pointwise convolution to integrate channel-wise information. Subsequently, a channel-shuffle operation is applied to facilitate information flow across feature groups by rearranging the tensor dimensions. Finally, a depthwise convolution performs per-channel local extraction; the preceding channel shuffle enables cross-group information communication before spatial–spectral filtering. The overall operation is formulated as

$$\text{Local}(\mathcal{X}) = \text{DWConv}_{3\times3\times3}\left(\text{Shuffle}\left(\text{Conv}_{1\times1\times1}(\mathbf{X})\right)\right) \tag{16}$$

where DWConv denotes the depthwise convolution and Shuffle$(\cdot)$ denotes the channel-shuffle operation.

4) SiLU–GRN Gated Feed-Forward Network: To further enhance nonlinear feature transformation, the SGG-FFN incorporates 3-D depthwise convolution, a SiLU-based gating mechanism, and GRN into the feed-forward transformation. Unlike standard FFNs, the proposed module incorporates 3-D depthwise convolutions within the high-dimensional feature space to capture local contextual information before the gating mechanism. As illustrated in Fig. 2(e), the input tensor of the SGG-FFN is the normalized intermediate tensor $\mathcal{Z}_{\text{in},l}$ defined in (6). It is first expanded along the channel dimension and encoded through pointwise and depthwise convolutions

$$\mathcal{F} = \text{DWConv}_{3\times3\times3}\left(\text{PWConv}_{1\times1\times1}\left(\mathcal{Z}_{\text{in},l}\right)\right) \tag{17}$$

where $\mathcal{F} \in \mathbb{R}^{B\times2C_h\times D\times H\times W}$ denotes the expanded intermediate feature tensor; C_h denotes the hidden channel dimension; and PWConv and DWConv denote pointwise and depthwise convolutions, respectively. The tensor $\mathcal{F}$ is evenly split along the channel dimension

$$[\mathcal{F}_1,\mathcal{F}_2] = \text{Split}_c(\mathcal{F}) \tag{18}$$

where $\mathcal{F}_1,\mathcal{F}_2 \in \mathbb{R}^{B\times C_h\times D\times H\times W}$. The first part is activated by the SiLU function and serves as a content-dependent gate for the second part

$$\mathcal{U} = \text{SiLU}(\mathcal{F}_1)\odot\mathcal{F}_2 \tag{19}$$

where $\odot$ denotes element-wise multiplication and $\mathcal{U} \in \mathbb{R}^{B\times C_h\times D\times H\times W}$ denotes the gated feature tensor. GRN first computes the global response over the spectral and spatial dimensions

$$\mathcal{G} = \|\mathcal{U}\|_{2,(D,H,W)} \in \mathbb{R}^{B\times C_h\times1\times1\times1} \tag{20}$$

followed by response normalization

$$\mathcal{R} = \frac{\mathcal{G}}{\max_c(\mathcal{G}) + \epsilon} \tag{21}$$

where $\max_c(\cdot)$ denotes the maximum response along the channel dimension and ϵ is a small constant for numerical stability. The GRN-enhanced feature tensor is defined as

$$\mathcal{F}_{\text{grn}} = \mathcal{U} + \boldsymbol{\gamma}_{\text{grn}}\odot(\mathcal{U}\odot\mathcal{R}) + \boldsymbol{\beta}_{\text{grn}} \tag{22}$$

where $\boldsymbol{\gamma}_{\text{grn}}$ and $\boldsymbol{\beta}_{\text{grn}}$ are learnable channel-wise scaling and bias parameters, respectively. Finally, a pointwise convolution projects the feature tensor back to the original channel dimension

$$\mathcal{Z}_{\text{out},l} = \text{PWConv}_{1\times1\times1}\left(\mathcal{F}_{\text{grn}}\right) \tag{23}$$

where $\mathcal{Z}_{\text{out},l} \in \mathbb{R}^{B\times C\times D\times H\times W}$ is the output tensor of the SGG-FFN. Therefore, SGG-FFN$(\mathcal{Z}_{\text{in},l}) = \mathcal{Z}_{\text{out},l}$, and the residual block output is given by (5).

C. Physics-Guided Unmixing Encoder II and Linear Reconstruction Decoder

The final stage of D3A-Net transforms the learned deep 3-D representations into physically meaningful unmixing results, strictly adhering to the constraints of the LMM. As shown in Fig. 1, this unmixing stage is composed of Encoder II and the linear Decoder. Encoder II maps the output feature tensor of Encoder I, denoted by $\mathcal{X}_{\text{E1}} \in \mathbb{R}^{B\times C\times D\times H\times W}$, to

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the 2-D abundance space. First, a Flatten operation collapses the channel (C) and spectral depth (D) dimensions into a single high-dimensional feature vector per pixel. Subsequently, a 1×1 convolution layer acts as a predictor to map these features to P endmembers. To enforce the physical constraints of the LMM, specifically the ANC and ASC, a Softmax function is applied along the endmember dimension. This operation explicitly guarantees that the predicted abundances are nonnegative and sum to one. Following the abundance estimation, we employ a linear decoder layer to reconstruct the HSI, where the weight matrix explicitly represents the endmember spectra and is constrained by an absolute value function to ensure nonnegativity.

D. Loss Function

The network is trained in an end-to-end unsupervised manner. To guide the optimization, we define a hybrid objective function that balances reconstruction fidelity with physical plausibility constraints on both spectra and abundances

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{MSE}} + \lambda_{\text{SAD}}\mathcal{L}_{\text{SAD}} + \lambda_{\text{sp}}\mathcal{L}_{\text{Sparsity}} \tag{24}$$

where λ_{SAD} and λ_{sp} denote the weighting coefficients of the spectral angle distance (SAD) loss and sparsity loss, respectively.

1) *Reconstruction Loss:* We employ the mean squared error (mse) to minimize the pixel-wise Euclidean distance between the observed HSI spectrum $\mathbf{y}_n$ and its reconstruction $\widehat{\mathbf{y}}_n$

$$\mathcal{L}_{\text{MSE}} = \frac{1}{N}\sum_{n=1}^{N} \|\mathbf{y}_n - \hat{\mathbf{y}}_n\|_2^2 \tag{25}$$

where $\mathbf{y}_n, \widehat{\mathbf{y}}_n \in \mathbb{R}^L$ denote the observed and reconstructed spectra of pixel n, respectively.

2) *SAD Loss:* While mse focuses on intensity values, spectral identity is largely encoded in the shape of the spectral curve. To ensure spectral shape consistency and mitigate the impact of scaling variations, we minimize the SAD error

$$\mathcal{L}_{\text{SAD}} = \frac{1}{N}\sum_{n=1}^{N} \arccos\left(\frac{\hat{\mathbf{y}}_n^\top \mathbf{y}_n}{\|\hat{\mathbf{y}}_n\|_2 \, \|\mathbf{y}_n\|_2}\right). \tag{26}$$

3) *Sparsity Loss:* In real-world scenarios, a mixed pixel is typically composed of only a small subset of the available endmembers. To encourage sparse solutions, we impose an $\ell_{1/2}$ sparsity constraint on the estimated abundance maps. The $\ell_{1/2}$ norm promotes sparsity more effectively than the ℓ_1 norm while remaining differentiable for backpropagation

$$\mathcal{L}_{\text{Sparsity}} = \frac{1}{N}\sum_{n=1}^{N} \|\hat{\mathbf{a}}_n\|_{1/2} = \frac{1}{N}\sum_{n=1}^{N}\sum_{p=1}^{P} \left(\hat{a}_{n,p} + \epsilon\right)^{1/2} \tag{27}$$

where $\widehat{\mathbf{a}}_n \in \mathbb{R}^P$ denotes the estimated abundance vector of pixel n, $\widehat{a}_{n,p}$ denotes its pth element, and ϵ is a small positive constant added for numerical stability.

IV. Experiments

Experiments were conducted to evaluate the proposed D3A-Net. We first describe the datasets, evaluation metrics, and implementation settings, and then compare D3A-Net with seven representative HU methods on one synthetic dataset and three real-world datasets. The comparison methods include one traditional model-based method, VCA+FCLS, as well as six DL-based methods, CyCU-Net [27], DFFN [30], SSAF-Net [29], A2SAN [31], DeepTrans [36], and MSNet [28].

A. Data Description

1) Synthetic Dataset: The synthetic dataset comprises 100×100 pixels across 224 spectral bands. This dataset was constructed based on the LMM described in [44], utilizing nine endmembers randomly selected from the United States Geological Survey (USGS) spectral library. The abundance coefficients of the synthetic dataset not only strictly satisfy the ANC and ASC constraints but also incorporate spatial smoothness and edge properties that closely approximate real-world ground features. The dataset was subsequently degraded by zero-mean Gaussian white noise with signal-to-noise ratios (SNRs) of 10, 20, and 30 dB.

2) Samson Dataset: The Samson dataset is a widely used benchmark for HU. The original image comprises 952×952 pixels with 156 spectral channels covering the wavelength range from 401 to 889 nm. A subimage of 95×95 pixels was cropped for experiments. It contains three primary endmembers: soil, tree, and water.

3) Jasper Ridge Dataset: This dataset was acquired by the Airborne Visible/Infrared Imaging Spectrometer (AVIRIS). The original scene consists of 512×614 pixels with 224 spectral bands ranging from 380 to 2500 nm. Consistent with previous works, we utilized a subscene of 100×100 pixels. After removing bands affected by water vapor and atmospheric effects (bands 1–3, 108–112, 154–166, and 220–224), 198 bands were retained. The scene mainly comprises four endmembers: road, soil, water, and tree.

4) APEX Dataset: The APEX dataset, obtained from the APEX sensor, is a complex scene suitable for evaluating the extraction of fine spatial structures. We selected a subimage of 110×110 pixels. The dataset contains four endmembers: road, tree, roof, and water. Notably, the road endmember in this scene exhibits a narrow, winding spatial distribution, which poses a significant challenge for fixed receptive-field networks.

B. Experimental Setup

1) Evaluation Metrics: To quantitatively assess the unmixing performance, we employ two standard metrics: the SAD and the root-mean-square error (RMSE). SAD measures the spectral shape similarity between the estimated endmember $\hat{\mathbf{m}}$ and the ground truth $\mathbf{m}$

$$\text{SAD} = \arccos\left(\frac{\hat{\mathbf{m}}^\top \mathbf{m}}{\|\hat{\mathbf{m}}\|_2 \, \|\mathbf{m}\|_2}\right). \tag{28}$$

RMSE evaluates the reconstruction error of the abundance maps $\hat{\mathbf{a}}$ compared to the ground truth $\mathbf{a}$

$$\text{RMSE} = \sqrt{\frac{1}{N}\sum\nolimits_{i=1}^{N} \|\hat{\mathbf{a}}_i - \mathbf{a}_i\|_2^2}. \tag{29}$$

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TABLE I
QUANTITATIVE COMPARISONS ON THE SYNTHETIC DATASET AT SNRS OF 10, 20, AND 30 DB

SNR	Metric	Method							
		VCA+FCLS	CyCU-Net	DFFN	SSAF-Net	A2SAN	DeepTrans	MSNet	Proposed
10 dB	Mean RMSE	0.1359	0.3069	0.1432	0.0968	0.1047	0.2603	0.0735	0.0635
	Mean SAD	0.1013	0.0458	0.1522	0.0922	0.1214	0.1709	0.0421	0.0220
20 dB	Mean RMSE	0.0512	0.2875	0.1467	0.0542	0.1081	0.3160	0.0691	0.0609
	Mean SAD	0.0380	0.0555	0.2785	0.0813	0.1199	0.1675	0.0345	0.0187
30 dB	Mean RMSE	0.0203	0.3210	0.1178	0.0646	0.1082	0.2040	0.0655	0.0545
	Mean SAD	0.0131	0.0549	0.1487	0.0841	0.1207	0.0916	0.0309	0.0166

TABLE II
QUANTITATIVE RESULTS OF MEAN RMSE AND MEAN SAD FOR THE SAMSON DATASET

Metric		Method							
		VCA+FCLS	CyCU-Net	DFFN	SSAF-Net	A2SAN	DeepTrans	MSNet	Proposed
SAD	Soil	0.0207	0.0365	0.0158	0.0348	0.0184	0.0347	0.0454	0.0409
	Tree	0.0495	0.1415	0.0313	0.0331	0.0391	0.0986	0.0180	0.0324
	Water	0.1299	0.0380	0.0518	0.0566	0.1240	0.1356	0.0756	0.0135
	Mean SAD	0.0667	0.0720	0.0426	0.0415	0.0613	0.0896	0.0331	0.0289
RMSE	Soil	0.2552	0.1916	0.0615	0.0713	0.0697	0.0768	0.0569	0.0821
	Tree	0.1807	0.1566	0.0533	0.0323	0.0445	0.0642	0.0621	0.0341
	Water	0.3488	0.1827	0.0300	0.0401	0.0557	0.0823	0.0271	0.0344
	Mean RMSE	0.2598	0.1776	0.0483	0.0479	0.0566	0.0748	0.0487	0.0502

TABLE III
QUANTITATIVE RESULTS OF MEAN RMSE AND MEAN SAD FOR THE JASPER RIDGE DATASET

Metric		Method							
		VCA+FCLS	CyCU-Net	DFFN	SSAF-Net	A2SAN	DeepTrans	MSNet	Proposed
SAD	Soil	0.0827	0.0974	0.0653	0.0466	0.0829	0.0979	0.0691	0.0395
	Tree	0.1036	0.1277	0.0802	0.1327	0.1099	0.1717	0.0284	0.0292
	Water	0.1521	0.0503	0.1102	0.0814	0.1355	0.1150	0.0859	0.0188
	Road	0.1980	0.0450	0.1152	0.0597	0.1573	0.1132	0.0363	0.0182
	Mean SAD	0.1341	0.0801	0.0927	0.0801	0.1214	0.0869	0.0549	0.0294
RMSE	Soil	0.2321	0.1257	0.0616	0.0467	0.0621	0.0915	0.1104	0.0723
	Tree	0.0736	0.1049	0.0325	0.0260	0.1168	0.0944	0.0602	0.0584
	Water	0.1861	0.1375	0.1018	0.0667	0.1057	0.1723	0.1410	0.0878
	Road	0.1671	0.0961	0.0812	0.1542	0.1422	0.1544	0.1226	0.0597
	Mean RMSE	0.1645	0.1172	0.0708	0.0734	0.1067	0.1331	0.1085	0.0691

Lower values for both SAD and RMSE indicate better performance.

2) Hyperparameter Settings: All experiments in this study were implemented using the PyTorch deep learning framework and executed on a high-performance workstation equipped with an NVIDIA GeForce RTX 2080 Ti GPU and an Intel Xeon E5-2620 v3 CPU. For the computationalefficiency evaluation, all methods were evaluated using the same Jasper Ridge scene and hardware platform The total unmixing time records the complete optimization and final-estimation procedure under a consistent timing boundary for all methods. Regarding the network architecture, the base-channel dimension was set to $C = 24$ to balance representation capacity and computational cost. The numbers of attention heads at the four hierarchical levels were set to $\{3, 6, 12, 24\}$. These values are divisible by the three-scale groups, enabling an equal assignment of attention heads while maintaining a consistent per-head channel dimension. The base neighborhood size and dilation rate were set to $3 \times 3 \times 3$ and 1, respectively, to preserve local spectral–spatial continuity, while the remaining scale groups enlarged the contextual range through increased dilation or neighborhood size. These architectural settings were fixed across all datasets. To optimize the network parameters, we employed the AdamW optimizer with a weight decay of 1×10^{-4}. The initial learning rate was set to 4×10^{-4}. To prevent training from stagnating in local minima and to ensure stable model convergence, we adopted the cosine annealing warm restarts scheduler, with an initial cycle (T_0)

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TABLE IV
QUANTITATIVE RESULTS OF MEAN RMSE AND MEAN SAD FOR THE APEX DATASET

Metric	Class	Method							
		VCA+FCLS	CyCU-Net	DFFN	SSAF-Net	A2SAN	DeepTrans	MSNet	Proposed
SAD	Road	0.0988	0.5719	0.2380	0.1501	0.1220	0.1819	0.1896	0.1165
	Tree	0.1285	0.1245	0.1434	0.1339	0.0430	0.1505	0.2850	0.1239
	Roof	0.0869	0.1307	0.1362	0.0991	0.1955	0.1415	0.1717	0.1083
	Water	0.1793	0.4359	0.1042	0.3450	0.1333	0.0990	0.0430	0.0793
	Mean SAD	0.1234	0.3158	0.1675	0.1820	0.1235	0.1507	0.1723	0.1070
RMSE	Road	0.1739	0.3615	0.1635	0.1212	0.1124	0.2475	0.2183	0.1362
	Tree	0.2162	0.1244	0.1646	0.1457	0.1673	0.1325	0.1885	0.1326
	Roof	0.2542	0.1031	0.1405	0.1857	0.1481	0.1537	0.1129	0.0948
	Water	0.4168	0.1066	0.1855	0.1754	0.1348	0.1168	0.2699	0.0501
	Mean RMSE	0.2653	0.1739	0.1635	0.1570	0.1410	0.1362	0.1974	0.1221

TABLE V
ABLATION STUDY OF KEY DESIGN CHOICES ON THE JASPER RIDGE AND APEX DATASETS (LOWER IS BETTER)

Design Choice							Jasper Ridge		APEX	
3D	MS-DNA	Multi-scale design	Deformable sampling	Local	GRN	Sparsity loss	Mean RMSE	Mean SAD	Mean RMSE	Mean SAD
×	✓	✓	✓	✓	✓	✓	0.1021	0.0320	0.1748	0.1227
✓	✓	✓	× (Fixed)	✓	✓	✓	0.0850	0.0362	0.1506	0.1310
✓	×	–	–	×	✓	✓	0.0939	0.0418	0.1653	0.1492
✓	✓	×	✓	✓	✓	✓	0.0769	0.0326	0.1367	0.1178
✓	✓	✓	✓	✓	×	✓	0.0745	0.0313	0.1319	0.1135
✓	✓	✓	✓	✓	✓	×	0.0798	0.0335	0.1410	0.1219
✓	✓	✓	✓	✓	✓	✓	**0.0691**	**0.0294**	**0.1221**	**0.1070**

TABLE VI
ABLATION STUDY OF KEY DESIGN CHOICES ON THE SYNTHETIC DATASET UNDER TWO REPRESENTATIVE NOISE LEVELS (LOWER IS BETTER)

Design Choice							10 dB		30 dB	
3D	MS-DNA	Multi-scale design	Deformable sampling	Local	GRN	Sparsity loss	Mean RMSE	Mean SAD	Mean RMSE	Mean SAD
×	✓	✓	✓	✓	✓	✓	0.0927	0.0254	0.0763	0.0185
✓	✓	✓	× (Fixed)	✓	✓	✓	0.0787	0.0271	0.0643	0.0196
✓	×	–	–	×	✓	✓	0.0864	0.0308	0.0709	0.0219
✓	✓	×	✓	✓	✓	✓	0.0711	0.0242	0.0594	0.0179
✓	✓	✓	✓	✓	×	✓	0.0686	0.0233	0.0572	0.0173
✓	✓	✓	✓	✓	✓	×	0.0737	0.0251	0.0610	0.0183
✓	✓	✓	✓	✓	✓	✓	**0.0635**	**0.0220**	**0.0545**	**0.0166**

of 50 epochs and a minimum learning rate of 1×10^{-6}. The total number of training epochs was set to 1000 for the Samson dataset and 2000 for the Jasper Ridge and APEX datasets. Regarding the objective function, the hyperparameters balancing the reconstruction loss, SAD loss, and sparsity loss were determined by the joint sensitivity analysis reported in Section IV-H. Specifically, the weight of the SAD term λ_{SAD} was set to 0.05, and the weight of the sparsity constraint λ_{sp} was set to 1×10^{-3}. Furthermore, to initialize the decoder weights, we utilized the VCA algorithm. This strategy provides a physically meaningful starting point, thereby effectively accelerating the convergence of the unmixing process.

3) Comparison Methods Settings: To validate the unmixing accuracy of D3A-Net, we compared it with seven representative HU methods, including one traditional geometry-based method, VCA+FCLS, and six DL-based methods: CyCU-Net, DFFN, SSAF-Net, A2SAN, DeepTrans, and MSNet. For all DL-based comparison methods, we followed the network configurations and parameter settings recommended in their original papers and performed dataset-specific parameter tuning when necessary. For methods requiring endmember initialization, VCA was consistently adopted to ensure a fair comparison. The above methods constitute the primary accuracy-comparison set used in Tables I–IV. The 3D-CNN-AE [45], by contrast, was originally developed for an additive nonlinear unmixing formulation, and its reconstruction objective and training formulation differ from those adopted in the primary benchmark comparison. Therefore, it is not included as an additional accuracy baseline in Tables I–IV. Nevertheless, because its dense native 3-D convolutions provide a relevant architectural reference for evaluating the computational cost of volumetric spectral–spatial processing, we additionally include 3D-CNN-AE in the computational-efficiency analysis of Section IV-I. Its results are used specifically to contextualize computational overhead rather than to extend the accuracy ranking reported in Tables I–IV.

C. Experiments on Synthetic Dataset

Table I shows that the advantage of D3A-Net is most pronounced under severe noise. At 10 dB, the proposed method achieves a mean RMSE of 0.0635 and a mean SAD of 0.0220, demonstrating stronger robustness than the comparison methods when the spectral–spatial structure is heavily corrupted. As the SNR increases, the performance gap gradually narrows, and VCA+FCLS obtains lower errors at 30 dB. This trend indicates that the primary advantage of D3A-Net lies in handling strongly perturbed observations rather than uniformly outperforming all methods under every noise level.

Fig. 3 compares the estimated spectra of representative endmembers, including EM1, EM3, EM5, EM7, and EM9, under the 30-dB condition. Several comparison methods exhibit visible fluctuations or excessive smoothing around

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TABLE VII

COMPUTATIONAL EFFICIENCY COMPARISON ON THE JASPER RIDGE DATASET. LOWER VALUES INDICATE LOWER COMPUTATIONAL COSTS

Method	Params (M)	FLOPs (G)	Total Unmixing Time (s)	Peak GPU Memory (MB)
VCA+FCLS	–	–	2.372	–
CyCU-Net	0.036	1.383	30.981	151.113
DFFN	1.869	14.705	29.603	45.632
SSAF-Net	0.665	12.555	87.083	216.387
A2SAN	0.043	0.889	75.177	156.458
DeepTrans	2.479	2.848	69.625	256.454
MSNet	0.653	5.732	28.767	186.072
3D-CNN-AE	6.353	133.660	357.502	545.506
Proposed	2.675	17.675	136.743	206.656

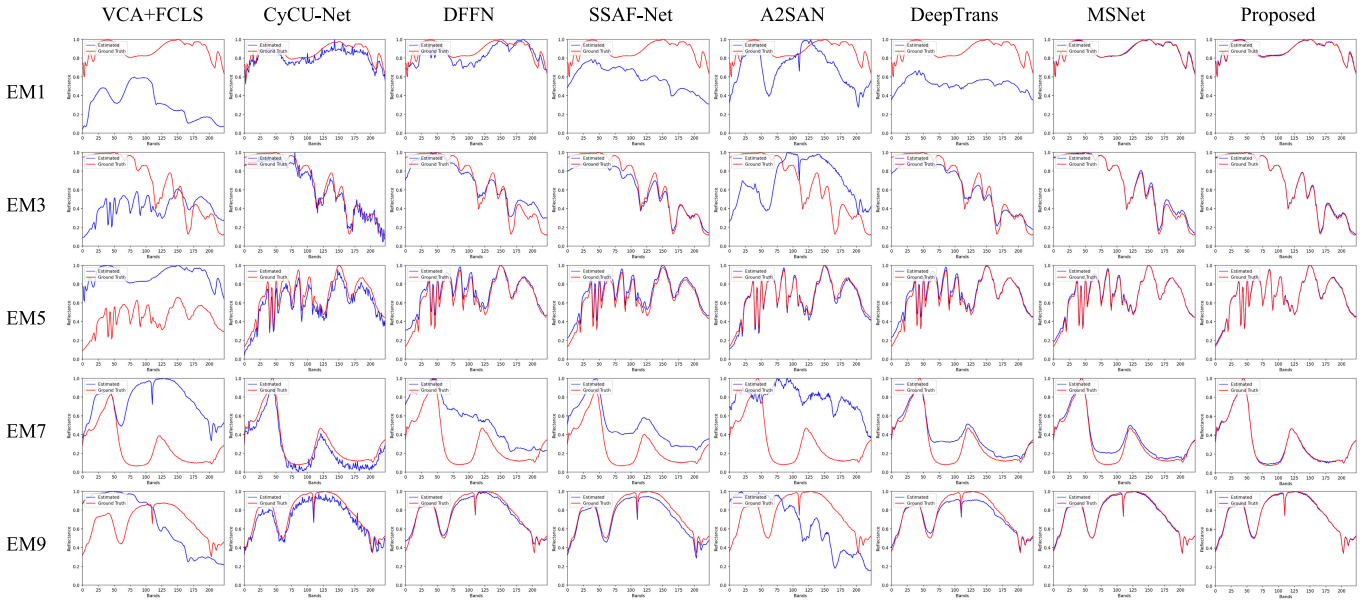


Fig. 3. Estimated endmembers by different methods (blue curves) and the corresponding ground truth (red curves) on the Synthetic dataset at 30-dB SNR.

local absorption peaks and valleys. In contrast, the spectra estimated by D3A-Net more closely follow the overall shapes and local variations of the reference spectra over the entire wavelength range. This behavior is associated with the native 3-D representation, which jointly models adjacent-band correlations and spatial context instead of treating individual pixel spectra independently. Consequently, random spectral variations that are inconsistent with the surrounding spectral–spatial structure have a reduced influence on the encoded features, thereby improving the preservation of endmember spectral shapes.

The abundance maps in Fig. 4 provide further spatial evidence. VCA+FCLS and CyCU-Net produce numerous isolated noisy pixels in the estimated abundance maps, particularly within homogeneous regions. DFFN alleviates part of the spatial discontinuity but still exhibits residual artifacts and diffuse abundance transitions. By comparison, D3A-Net reconstructs abundance maps with smoother distributions inside homogeneous regions, fewer spurious abundance estimates, and better-preserved transitions between different materials.

Mechanistically, MS-DNA aggregates features over content-adaptive 3-D neighborhoods, enabling the model to emphasize sampled features that are spatially and spectrally consistent with the target pixel while reducing the influence of sampled features dominated by noise. The complementary local extraction branch preserves short-range continuity and local abundance variations that may otherwise be weakened during adaptive aggregation. In addition, the SAD loss constrains the angular consistency and spectral shape of the reconstructed pixels, whereas the sparsity constraint suppresses diffuse low-magnitude abundance estimates. The interaction among native 3-D representation, adaptive neighborhood aggregation, local feature preservation, and physically motivated losses explains the improved stability of D3A-Net under low-SNR conditions.

D. Experiments on Samson Dataset

Table II reports the quantitative results on the Samson dataset. D3A-Net achieves the lowest mean SAD of 0.0289, indicating superior preservation of material-specific spectral shapes. Its mean RMSE is 0.0502, which is slightly higher than those of several comparison methods. The different rankings of the two metrics arise from their distinct error sensitivities. SAD primarily evaluates the angular difference between the estimated and reference endmembers, whereas abundance

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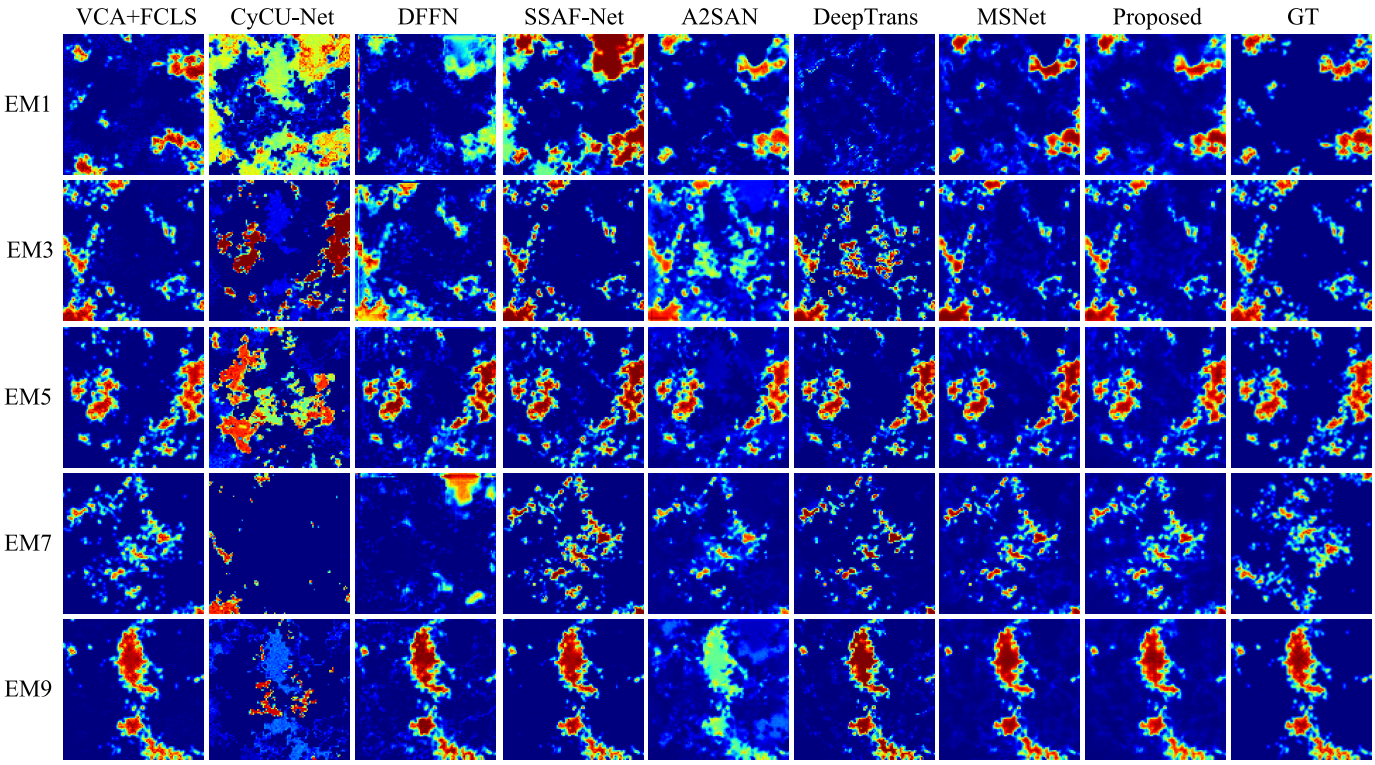


Fig. 4. Estimated abundance maps for different endmembers on the Synthetic dataset at 30-dB SNR.

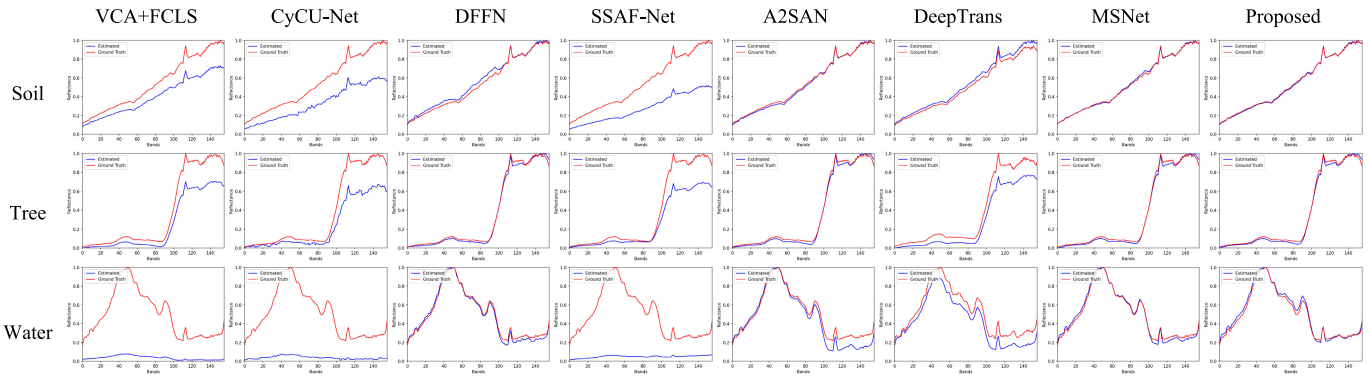


Fig. 5. Estimated endmembers by different methods (blue curves) and the corresponding ground truth (red curves) on the Samson dataset.

RMSE measures pixel-wise deviations in the estimated material proportions. Therefore, the Samson results indicate that D3A-Net provides a stronger advantage in endmember spectral estimation than in pixel-wise abundance estimation.

The estimated endmember spectra in Fig. 5 support this observation. The endmembers estimated by D3A-Net closely follow the reference spectra and preserve the principal absorption and reflection characteristics, with relatively small deviations in local spectral shape. The abundance maps in Fig. 6 further reveal the differences in spatial reconstruction. Some comparison methods produce spurious nonzero abundance estimates for other endmembers within regions dominated by a single material. Diffuse abundance transitions and abundance leakage can also be observed near the boundaries between soil, tree, and water. In contrast, D3A-Net produces smoother abundance distributions within homogeneous regions while preserving spatially localized transitions between different materials.

This behavior results from the complementary functions of the two feature-extraction branches. The MS-DNA branch aggregates repeated spectral–spatial evidence over adaptive neighborhoods, thereby suppressing isolated abundance estimates that are inconsistent with the dominant material distribution. Meanwhile, the local extraction branch and skip connections retain short-range contrast and fine spatial details near material transitions. Since Samson mainly contains relatively large homogeneous regions and comparatively regular material boundaries, the geometric advantage of deformable sampling is less pronounced than on Jasper Ridge and APEX. This observation is consistent with the relatively small RMSE differences among the leading methods in Table II.

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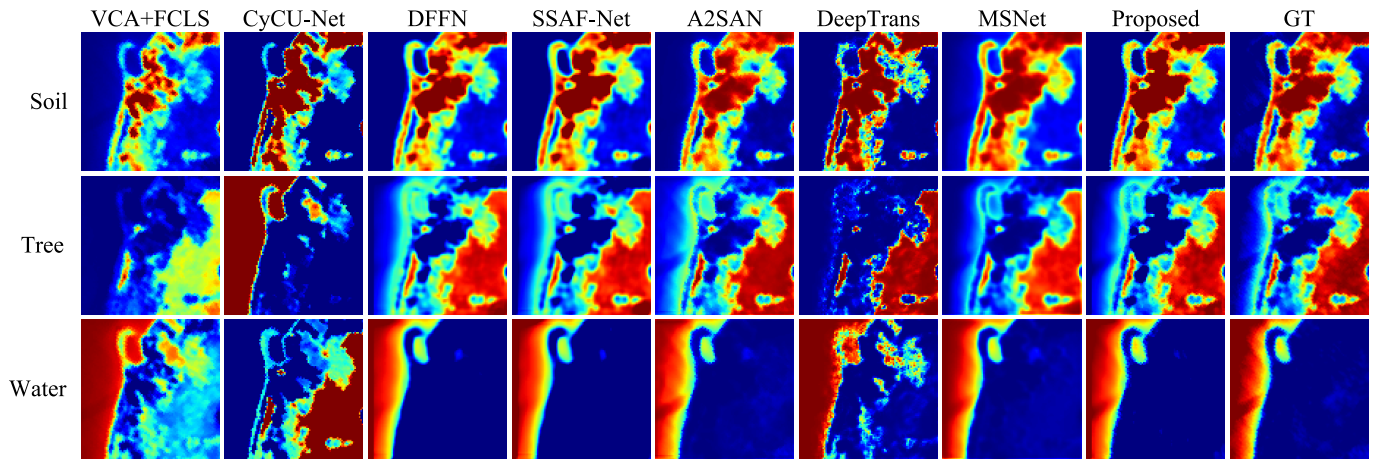


Fig. 6. Estimated abundance maps for different endmembers on the Samson dataset.

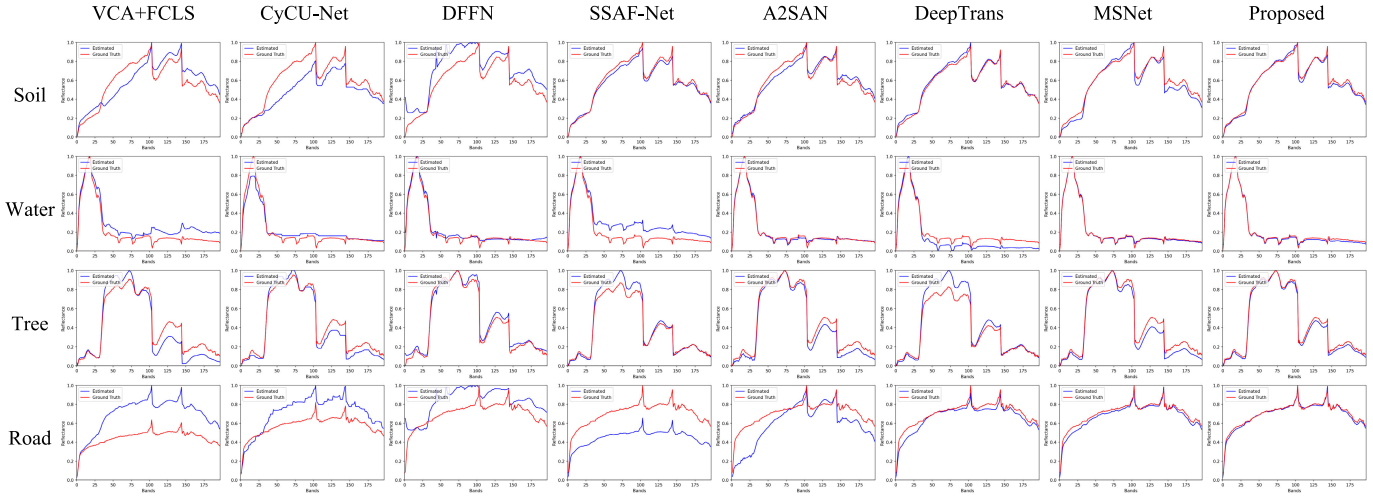


Fig. 7. Estimated endmembers by different methods (blue curves) and the corresponding ground truth (red curves) on the Jasper Ridge dataset.

E. Experiments on Jasper Ridge Dataset

Compared with Samson, Jasper Ridge contains more irregular spatial distributions, among which road is particularly challenging because of its narrow and elongated geometry. As reported in Table III, D3A-Net achieves the best mean SAD and mean RMSE. For the road endmember, it obtains an SAD of 0.0182 and an RMSE of 0.0597, demonstrating a clear advantage in reconstructing materials with irregular spatial distributions.

The estimated endmember spectra in Fig. 7 show that D3A-Net maintains close agreement with the reference spectra, particularly for the road endmember. The abundance maps in Fig. 8 provide more direct spatial evidence. The road abundance maps produced by several comparison methods contain discontinuous segments, diffuse boundaries, or spurious road abundance estimates in adjacent soil and vegetation regions. By contrast, D3A-Net reconstructs a more continuous road abundance distribution with better-defined spatial boundaries and substantially reduced interference from neighboring materials.

To directly examine the adaptive sampling behavior of MS-DNA, Fig. 9 visualizes the learned 3-D offsets on the Jasper Ridge scene. Fig. 9(a) overlays the spatial-offset vectors on the reference road abundance map and enlarges a representative region to show the sampling displacements of the three-scale groups. The gray markers denote the regular reference centers, whereas the colored markers denote the displaced sampling positions associated with the base, dilated, and larger kernel groups. The vectors are magnified by a factor of eight only for visual clarity. The reference abundance map is used only for spatial comparison and is not involved in offset prediction or postprocessing. Fig. 9(b) and (c) further presents the mean spatial- and spectral-offset magnitudes, respectively, with the white curves indicating the contours of the reference road abundance.

As shown in Fig. 9(b), relatively large spatial offsets are distributed along the narrow road structure and its surrounding transition regions, whereas most homogeneous background regions exhibit smaller responses. The enlarged region in Fig. 9(a) further shows that the displacement directions vary across spatial locations and scale groups rather than following a uniform fixed pattern. Meanwhile, the spectral offsets in Fig. 9(c) exhibit a related spatial distribution but remain bounded below approximately 0.07 voxel. This comparatively conservative adjustment along the spectral dimension is consistent with preserving local spectral continuity while adapting the spatial sampling positions to complex material geometry.

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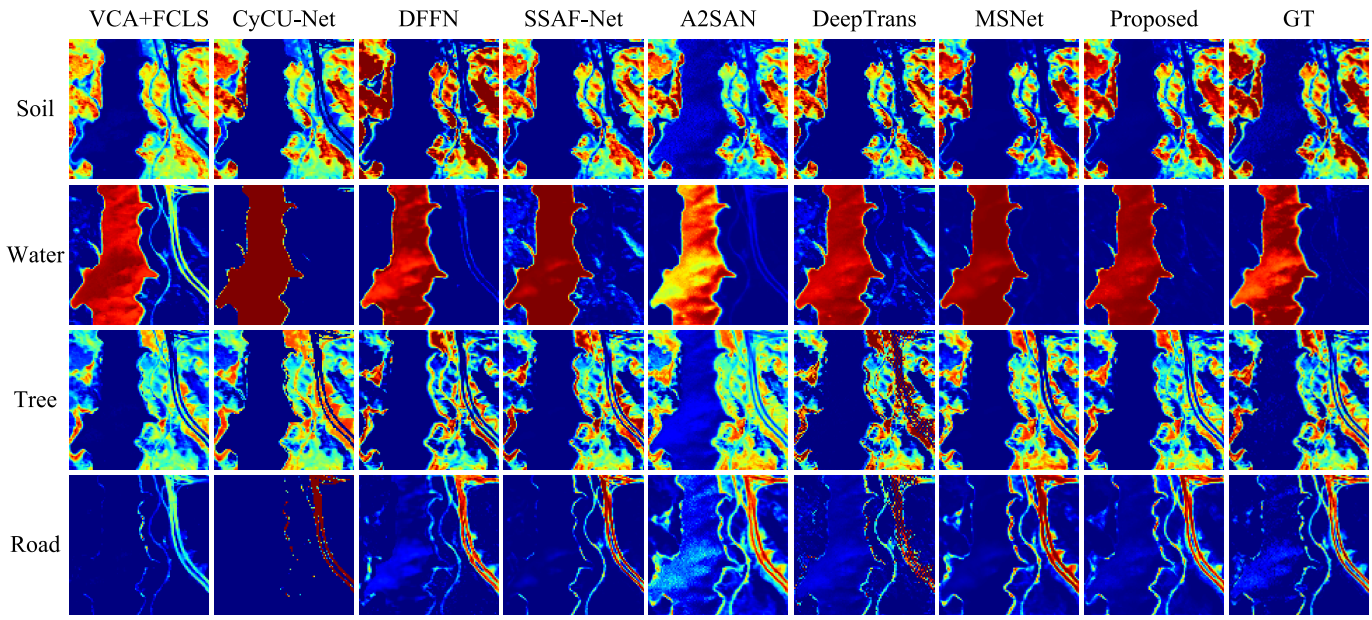


Fig. 8. Estimated abundance maps for different endmembers on the Jasper Ridge dataset.

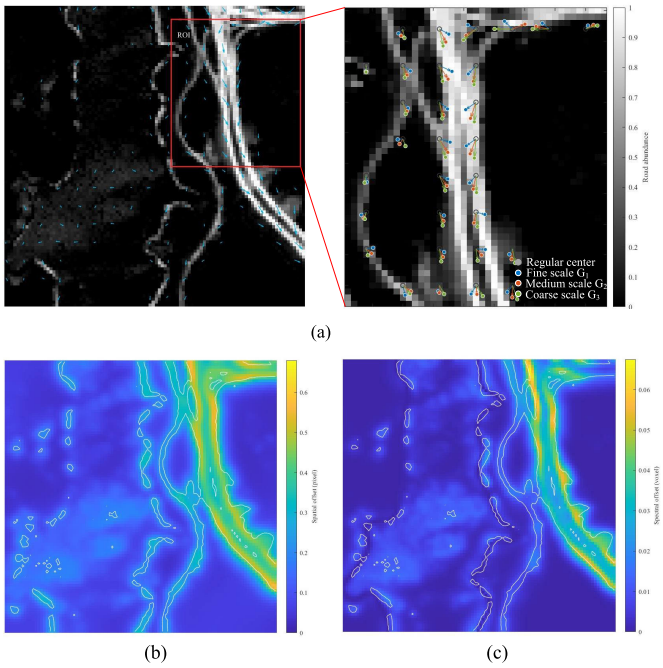


Fig. 9. Learned 3-D offsets of MS-DNA on Jasper Ridge. (a) Spatial offsets overlaid on the reference road abundance map, with an enlarged view; gray markers denote regular centers, while blue, orange, and green markers denote displaced positions for groups G_1–G_3, respectively. Vectors are enlarged by 8× for visibility. (b) Mean spatial-offset magnitude. (c) Mean spectral-offset magnitude.

These observations provide qualitative evidence of content-dependent 3-D sampling and should not be interpreted as a direct measurement of unmixing accuracy.

The visualization is further supported by the fixed-versus-deformable comparison in Table V. Replacing deformable sampling with fixed sampling increases the mean RMSE from 0.0691 to 0.0850 and the mean SAD from 0.0294 to 0.0362. Therefore, the learned-offset visualization and the controlled ablation provide complementary qualitative and quantitative evidence for the contribution of adaptive sampling to the reconstruction of narrow and irregular structures.

F. Experiments on APEX Dataset

Table IV reports the quantitative results on the APEX dataset. D3A-Net achieves the best overall performance, with a mean SAD of 0.1070 and a mean RMSE of 0.1221. Nevertheless, its advantage is class-dependent rather than uniform across all endmembers. The most evident improvements are observed in the abundance RMSEs of roof and water, whereas several comparison methods remain competitive for individual road- and tree-related metrics. Therefore, the APEX results indicate that D3A-Net achieves a better overall balance between endmember estimation and abundance estimation rather than class-wise dominance for every material.

The estimated endmember spectra in Fig. 10 generally follow the reference spectral shapes and retain the principal absorption and reflection characteristics. For tree, D3A-Net reconstructs the major near-infrared reflectance pattern, although the quantitative results show that other methods remain superior for some tree-related metrics. This difference is reasonable because the current linear decoder represents each material using one global endmember spectrum and does not explicitly model pixel-wise EV caused by illumination, canopy composition, or local environmental conditions.

Fig. 11 further compares the estimated abundance maps. Several comparison methods produce fragmented road abundance estimates, diffuse roof boundaries, or spurious abundance estimates of one endmember within regions dominated by another. In contrast, D3A-Net produces a more continuous road abundance map, better-defined spatial boundaries for the roof abundance distribution, and fewer spurious water abundance estimates outside regions dominated by water.

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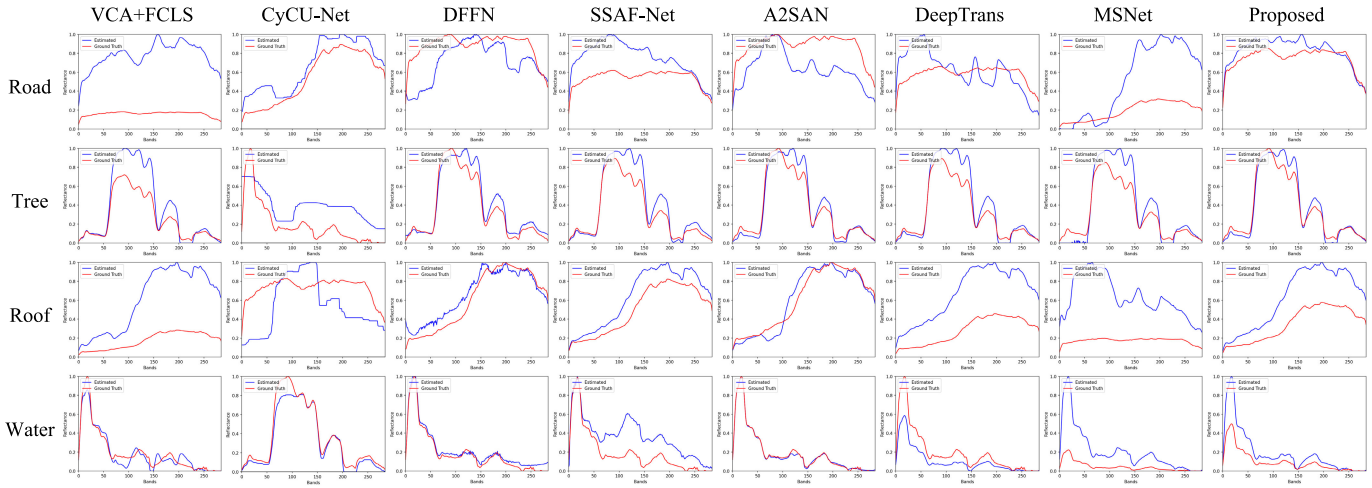


Fig. 10. Estimated endmembers by different methods (blue curves) and the corresponding ground truth (red curves) on the APEX dataset.

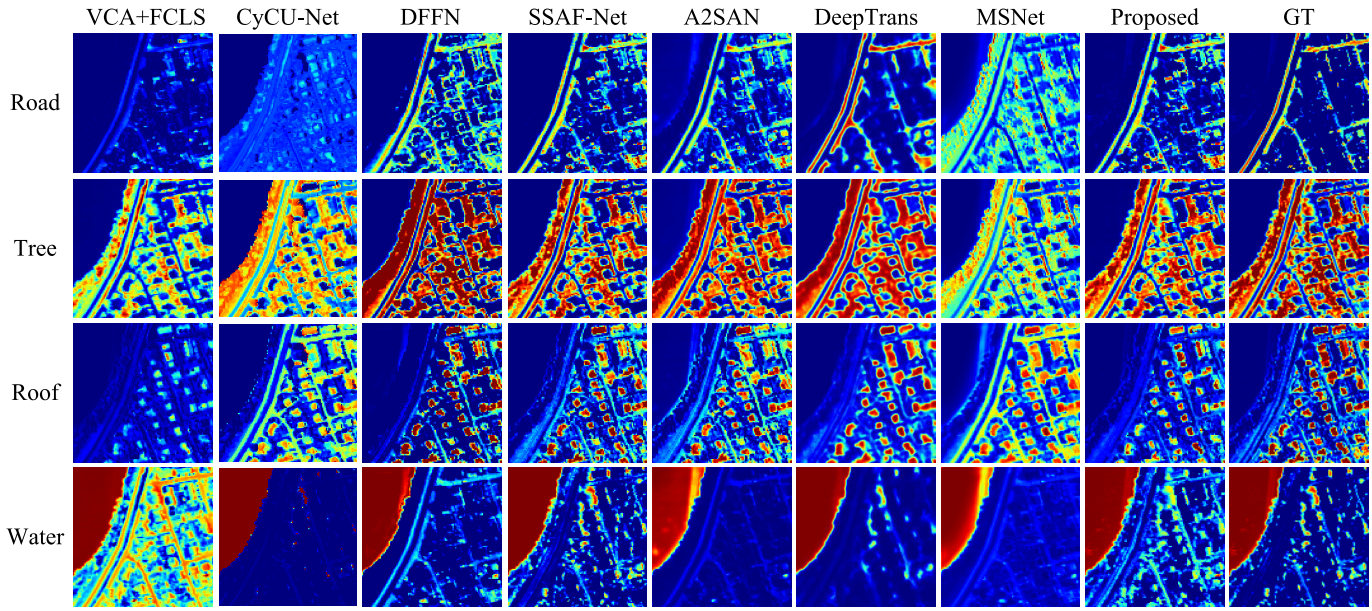


Fig. 11. Estimated abundance maps for different endmembers on the APEX dataset.

These visual improvements are consistent with the lower mean RMSE reported in Table IV.

Mechanistically, the irregular urban structures in APEX require the network to distinguish narrow or fragmented target regions from spectrally dissimilar neighboring materials. The learned offsets in MS-DNA adapt the sampling grid to the local material geometry, allowing attention aggregation to follow elongated road structures and irregular material boundaries while avoiding irrelevant neighboring regions.

The multiscale attention heads further combine fine-scale boundary information with broader contextual evidence. Heads with smaller neighborhoods preserve local abundance transitions, whereas heads with larger or dilated neighborhoods capture spatially separated but spectrally consistent regions. The local extraction branch complements this adaptive aggregation by retaining short-range continuity, local texture information, and fine material boundaries.

Consequently, D3A-Net produces more spatially coherent abundance estimates for materials with complex geometric distributions. At the same time, the tree results reveal an important limitation of the current framework. Geometry-adaptive feature aggregation cannot completely replace explicit modeling of EV. The APEX experiment therefore demonstrates both the effectiveness of the proposed geometric adaptation and the boundary of its applicability.

G. Ablation Study

To assess whether the contributions of the key design choices remain consistent across scene types and noise conditions, componentwise ablations are conducted on the Jasper Ridge and APEX datasets in Table V and on the synthetic dataset at 10 and 30 dB in Table VI. The two synthetic settings correspond to the endpoints of the evaluated SNR range. The variants are defined by the design choices listed in the tables: a check mark indicates that a component is enabled, a cross indicates that it is disabled, and a dash denotes an inapplicable option. The fixed-sampling variant retains the 3-D representation and multiscale neighborhoods but disables the learned

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offsets, whereas disabling the multiscale design replaces the three-scale aggregation with its single-scale counterpart.

Among the evaluated ablation variants, the complete configuration achieves the lowest mean RMSE and mean SAD: 0.0691/0.0294 on Jasper Ridge, 0.1221/0.1070 on APEX, 0.0635/0.0220 at 10 dB, and 0.0545/0.0166 at 30 dB. Replacing deformable sampling with fixed sampling increases the corresponding metric pairs to 0.0850/0.0362, 0.1506/0.1310, 0.0787/0.0271, and 0.0643/0.0196. The non-3-D variant and the configuration jointly removing MS-DNA and the local branch also produce higher errors in all four settings. Because MS-DNA and the local branch are removed simultaneously in the latter variant, this row evaluates only their joint global–local contribution and is not interpreted as an independent measurement of the local branch.

The individual ablations further clarify the contributions of the multiscale design, GRN, and sparsity loss. On Jasper Ridge, removing these components increases the mean RMSE/mean SAD from 0.0691/0.0294 to 0.0769/0.0326, 0.0745/0.0313, and 0.0798/0.0335. On APEX, the corresponding metric pairs increase from 0.1221/0.1070 to 0.1367/0.1178, 0.1319/0.1135, and 0.1410/0.1219. The same trend is observed on the synthetic dataset at both SNR levels, where every corresponding ablation yields higher mean RMSE and mean SAD than the complete configuration. Moreover, the absolute degradations caused by removing each of these three components are larger at 10 dB than at 30 dB, suggesting that their contributions become more pronounced under severe noise.

Mechanistically, the multiscale head groups combine fine-scale material transitions with broader spectral–spatial context, whereas GRN regulates channel-wise response magnitudes within the SGG-FFN and can reduce the dominance of a small subset of feature channels. The sparsity loss discourages diffuse low-magnitude abundance coefficients, consistent with the assumption that each mixed pixel generally contains only a subset of the available endmembers. Among these three individual ablations, removing the sparsity loss produces the largest numerical degradation in all four settings. Collectively, Tables V and VI indicate that these components provide complementary contributions rather than improvements restricted to the Jasper Ridge scene.

H. Hyperparameter Sensitivity Analysis

To examine the robustness of D3A-Net to the two task-specific loss weights, we conducted a joint sensitivity analysis on the Jasper Ridge dataset while keeping all other network and training settings unchanged. Jasper Ridge was selected as a representative challenging scene because it contains multiple materials with irregular spatial distributions. Specifically, λ_{SAD} was varied over $\{0.01, 0.02, 0.03, 0.04, 0.05, 0.075, 0.1\}$, while λ_{sp} was varied over $\{10^{-4}, 2\times10^{-4}, 4\times10^{-4}, 7\times10^{-4}, 10^{-3}, 3\times10^{-3}, 10^{-2}\}$.

As shown in Fig. 12, insufficient λ_{SAD} weakens the spectral-shape constraint, whereas an excessively large value may overemphasize angular consistency relative to abundance reconstruction. Similarly, a very small λ_{sp} provides insufficient suppression of diffuse abundance responses, while an excessively strong sparsity constraint may eliminate valid low-abundance components in highly mixed pixels. Among all tested combinations, $(\lambda_{\mathrm{SAD}}, \lambda_{\mathrm{sp}}) = (0.05, 10^{-3})$ achieves the lowest mean RMSE of 0.0691 and the lowest mean SAD of 0.0294.

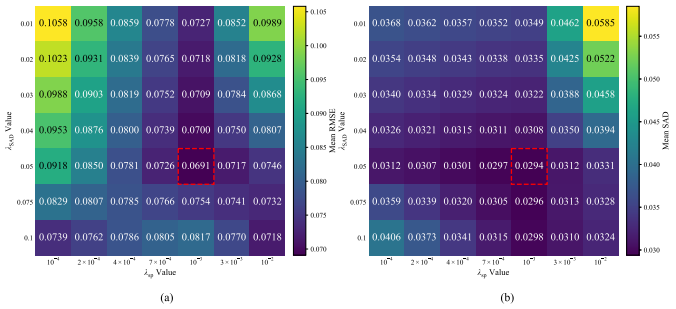


Fig. 12. Joint sensitivity analysis of D3A-Net with respect to the loss weights on the Jasper Ridge dataset. (a) RMSE. (b) SAD.

More importantly, a relatively broad low-error region is observed around the selected configuration. For $\lambda_{\mathrm{SAD}} \in [0.04, 0.075]$ and $\lambda_{\mathrm{sp}} \in [7\times10^{-4}, 10^{-3}]$, the mean RMSE remains within $[0.0691, 0.0766]$, while the mean SAD remains within $[0.0294, 0.0311]$. Therefore, the selected configuration does not correspond to an isolated optimum, indicating that D3A-Net is reasonably stable with respect to moderate variations in the two loss weights. Accordingly, $\lambda_{\mathrm{SAD}} = 0.05$ and $\lambda_{\mathrm{sp}} = 10^{-3}$ are adopted throughout all experiments.

I. Computational Efficiency Analysis

Following the comparison protocol described in Section IV-B, Table VII reports the computational costs of the primary comparison methods on the Jasper Ridge dataset and additionally includes 3D-CNN-AE as a structural efficiency reference. This additional baseline is introduced specifically to examine whether native 3-D spectral–spatial modeling necessarily incurs the high computational cost associated with dense 3-D convolution; it is not used to extend the accuracy rankings reported in Tables I–IV. All methods were evaluated using the same Jasper Ridge scene and hardware platform under a consistent timing boundary. CyCU-Net and A2SAN contain relatively few learnable parameters, whereas DFFN and SSAF-Net require larger numbers of operations because their feature extraction and reconstruction are performed at the full spatial resolution. DeepTrans exhibits moderate FLOPs but relatively high peak memory, indicating that parameter count, arithmetic complexity, and activation memory do not necessarily follow the same trend. Moreover, the total unmixing time is not strictly proportional to FLOPs because it is also affected by the training schedule, preprocessing operations, operator implementation, and GPU parallelism.

D3A-Net is more computationally demanding than several lightweight 2-D or pseudo-3D methods because it retains native volumetric representations and additionally performs deformable-offset prediction and multiscale neighborhood aggregation. Thus, D3A-Net is not designed to minimize computational cost across all model categories; instead, it targets an accuracy–efficiency tradeoff while preserving

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3-D spectral–spatial continuity. In comparison with the dense patch-wise 3D-CNN-AE, D3A-Net reduces repeated local-volume computation through bounded-neighborhood attention, lightweight pointwise/depthwise 3-D operations, and progressive spatial downsampling. D3A-Net requires fewer parameters, FLOPs, total unmixing time, and peak GPU memory than 3D-CNN-AE. Therefore, the efficiency advantage of D3A-Net should be interpreted relative to conventional dense 3D-CNN-based HU architectures, while its higher cost than lightweight 2-D methods represents a deliberate tradeoff for preserving native spectral–spatial continuity and improving unmixing accuracy in complex scenes.

V. Conclusion

This article presented D3A-Net, a 3-D deformable-attention framework for HU, designed to improve spectral–spatial representation learning under complex real-scene geometries. By processing hyperspectral imagery as a native 3-D cube within a U-shaped encoder–decoder, the proposed network preserves volumetric spectral–spatial continuity while enabling multiscale feature aggregation. In particular, D3A-Net introduces the 3D-DFB, where 3-D MS-DNA dynamically adjusts sampling locations to better align with irregular and slender structures, thereby mitigating background-induced feature interference and enhancing the fidelity of abundance estimation. The parallel combination of multiscale deformable attention and a lightweight 3-D local convolution branch further balances nonlocal dependency modeling with fine texture and boundary recovery, leading to sharper abundance maps and improved spectral reconstruction quality. Comprehensive comparisons and componentwise ablations on both synthetic and real benchmark datasets support the effectiveness of the proposed design across scene types and noise conditions. Within the ablation experiments, the complete configuration achieves the lowest mean RMSE and mean SAD on Jasper Ridge, APEX, and the synthetic dataset at both 10 and 30 dB, while removing the multiscale design, GRN, or sparsity loss consistently degrades both metrics. On real datasets, D3A-Net achieved state-of-the-art performance, including the best average SAD on Samson and Jasper Ridge, and the best overall results on APEX, demonstrating its practical advantages in scenes containing complex boundaries and heterogeneous materials. Together, the fixed-versus-deformable ablation and the learned-offset visualization provide complementary quantitative and qualitative evidence for the contribution of adaptive sampling, particularly for fine-grained structures that are difficult to represent using rigid-grid feature extraction. Future work will focus on adapting D3A-Net to more challenging unmixing scenarios, including EV and nonlinear mixing, while further improving its computational efficiency.

References

[1] A. F. H. Goetz, G. Vane, J. E. Solomon, and B. N. Rock, "Imaging spectrometry for Earth remote sensing," *Science*, vol. 228, no. 4704, pp. 1147–1153, Jun. 1985.

[2] J. M. Bioucas-Dias, A. Plaza, G. Camps-Valls, P. Scheunders, N. Nasrabadi, and J. Chanussot, "Hyperspectral remote sensing data analysis and future challenges," *IEEE Geosci. Remote Sens. Mag.*, vol. 1, no. 2, pp. 6–36, Jun. 2013.

[3] P. S. Thenkabail, M. K. Gumma, P. G. Teluguntla, and M. Ilyas, "Hyperspectral remote sensing of vegetation and agricultural crops," *Photogramm. Eng. Remote Sens.*, vol. 80, no. 4, pp. 697–709, 2014.

[4] F. D. van der Meer et al., "Multi- and hyperspectral geologic remote sensing: A review," *Int. J. Appl. Earth Observ. Geoinf.*, vol. 14, no. 1, pp. 112–128, 2012.

[5] G. A. Blackburn, "Hyperspectral remote sensing of plant pigments," *J. Experim. Botany*, vol. 58, no. 4, pp. 855–867, Nov. 2006.

[6] X. Tong, H. Xie, and Q. Weng, "Urban land cover classification with airborne hyperspectral data: What features to use?" *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 7, no. 10, pp. 3998–4009, Oct. 2014.

[7] D. A. Landgrebe, "Hyperspectral image data analysis," *IEEE Signal Process. Mag.*, vol. 19, no. 1, pp. 17–28, Jan. 2002.

[8] N. Keshava and J. F. Mustard, "Spectral unmixing," *IEEE Signal Process. Mag.*, vol. 19, no. 1, pp. 44–57, Jan. 2002.

[9] J. M. Bioucas-Dias et al., "Hyperspectral unmixing overview: Geometrical, statistical, and sparse regression-based approaches," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 5, no. 2, pp. 354–379, Apr. 2012.

[10] A. Plaza et al., "Recent advances in techniques for hyperspectral image processing," *Remote Sens. Environ.*, vol. 113, no. 1, pp. S110–S122, 2009.

[11] W.-K. Ma et al., "A signal processing perspective on hyperspectral unmixing: Insights from remote sensing," *IEEE Signal Process. Mag.*, vol. 31, no. 1, pp. 67–81, Jan. 2014.

[12] D. C. Heinz and Chein-I-Chang, "Fully constrained least squares linear spectral mixture analysis method for material quantification in hyperspectral imagery," *IEEE Trans. Geosci. Remote Sens.*, vol. 39, no. 3, pp. 529–545, Mar. 2001.

[13] M. E. Winter, "N-FINDR: An algorithm for fast autonomous spectral end-member determination in hyperspectral data," *Proc. SPIE*, vol. 3753, pp. 266–275, Jan. 1999.

[14] J. M. P. Nascimento and J. M. B. Dias, "Vertex component analysis: A fast algorithm to unmix hyperspectral data," *IEEE Trans. Geosci. Remote Sens.*, vol. 43, no. 4, pp. 898–910, Apr. 2005.

[15] J. Li and J. M. Bioucas-Dias, "Minimum volume simplex analysis: A fast algorithm to unmix hyperspectral data," in *Proc. IEEE Int. Geosci. Remote Sens. Symp.*, Jul. 2008, pp. III-250–III-253.

[16] J. M. P. Nascimento and J. M. B. Dias, "Does independent component analysis play a role in unmixing hyperspectral data?" *IEEE Trans. Geosci. Remote Sens.*, vol. 43, no. 1, pp. 175–187, Jan. 2005.

[17] M. Berman, H. Kiiveri, R. Lagerstrom, A. Ernst, R. Dunne, and J. F. Huntington, "ICE: A statistical approach to identifying endmembers in hyperspectral images," *IEEE Trans. Geosci. Remote Sens.*, vol. 42, no. 10, pp. 2085–2095, Oct. 2004.

[18] J. M. Bioucas-Dias and M. A. T. Figueiredo, "Alternating direction algorithms for constrained sparse regression: Application to hyperspectral unmixing," in *Proc. 2nd Workshop Hyperspectral Image Signal Process., Evol. Remote Sens.*, Jun. 2010, pp. 1–4.

[19] M.-D. Iordache, J. M. Bioucas-Dias, and A. Plaza, "Sparse unmixing of hyperspectral data," *IEEE Trans. Geosci. Remote Sens.*, vol. 49, no. 6, pp. 2014–2039, Jun. 2011.

[20] G. Martin and A. Plaza, "Spatial–spectral preprocessing prior to endmember identification and unmixing of remotely sensed hyperspectral data," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 5, no. 2, pp. 380–395, Apr. 2012.

[21] A. Plaza, P. Martinez, R. Perez, and J. Plaza, "Spatial/spectral endmember extraction by multidimensional morphological operations," *IEEE Trans. Geosci. Remote Sens.*, vol. 40, no. 9, pp. 2025–2041, Sep. 2002.

[22] B. Palsson, J. Sigurdsson, J. R. Sveinsson, and M. O. Ulfarsson, "Hyperspectral unmixing using a neural network autoencoder," *IEEE Access*, vol. 6, pp. 25646–25656, 2018.

[23] J. S. Bhatt and M. V. Joshi, "Deep learning in hyperspectral unmixing: A review," in *Proc. IEEE Int. Geosci. Remote Sens. Symp.*, Sep. 2020, pp. 2189–2192.

[24] Y. Su, J. Li, A. Plaza, A. Marinoni, P. Gamba, and S. Chakravortty, "DAEN: Deep autoencoder networks for hyperspectral unmixing," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 7, pp. 4309–4321, Jul. 2019.

[25] L. Wan, T. Chen, A. Plaza, and H. Cai, "Hyperspectral unmixing based on spectral and sparse deep convolutional neural networks," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 14, pp. 11669–11682, 2021.

[26] B. Palsson, M. O. Ulfarsson, and J. R. Sveinsson, "Convolutional autoencoder for spectral–spatial hyperspectral unmixing," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 1, pp. 535–549, Jan. 2021.

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[27] L. Gao, Z. Han, D. Hong, B. Zhang, and J. Chanussot, "CyCU-Net: Cycle-consistency unmixing network by learning cascaded autoencoders," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5503914.
[28] Y. Yu, Y. Ma, X. Mei, F. Fan, J. Huang, and H. Li, "Multi-stage convolutional autoencoder network for hyperspectral unmixing," *Int. J. Appl. Earth Observ. Geoinf.*, vol. 113, Sep. 2022, Art. no. 102981.
[29] W. Gao, J. Yang, Y. Zhang, Y. Akoudad, and J. Chen, "SSAF-Net: A spatial–spectral adaptive fusion network for hyperspectral unmixing with endmember variability," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, 2025, Art. no. 5506015.
[30] X. Tao, B. Koirala, A. Plaza, and P. Scheunders, "A new dual-feature fusion network for enhanced hyperspectral unmixing," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5540113.
[31] X. Tao, M. E. Paoletti, Z. Wu, J. M. Haut, P. Ren, and A. Plaza, "An abundance-guided attention network for hyperspectral unmixing," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5505414.
[32] L. Qi, M. Yue, F. Gao, B. Cao, J. Dong, and X. Gao, "Deep attention-guided spatial–spectral network for hyperspectral image unmixing," *IEEE Geosci. Remote Sens. Lett.*, vol. 21, pp. 1–5, 2024.
[33] L. Qi, Y. Wu, F. Gao, J. Dong, Q. Du, and X. Gao, "MCDB-Net: Multiview collaborative dual-branch unmixing network for hyperspectral images," *IEEE Trans. Geosci. Remote Sens.*, vol. 64, 2026, Art. no. 5507716.
[34] Y. Li, H. Zhang, and Q. Shen, "Spectral–spatial classification of hyperspectral imagery with 3D convolutional neural network," *Remote Sens.*, vol. 9, no. 1, p. 67, Jan. 2017.
[35] D. Hong et al., "SpectralFormer: Rethinking hyperspectral image classification with transformers," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5518615.
[36] P. Ghosh, S. K. Roy, B. Koirala, B. Rasti, and P. Scheunders, "Hyperspectral unmixing using transformer network," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5535116.
[37] Y. Su, L. Gao, A. Plaza, X. Sun, M. Jiang, and G. Yang, "SRViT: Self-supervised relation-aware vision transformer for hyperspectral unmixing," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 36, no. 10, pp. 17585–17598, Oct. 2025.
[38] Y. Zhang, J. Huang, Y. Huang, W. Gao, and J. Chen, "SSST-GAN: A sampling-based spatial–spectral transformer and generative adversarial network for hyperspectral unmixing," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 19, pp. 5741–5757, 2026.
[39] A. Dosovitskiy et al., "An image is worth 16×16 words: Transformers for image recognition at scale," in *Proc. Int. Conf. Learn. Represent. (ICLR)*, 2021, pp. 1–22.
[40] J. Dai et al., "Deformable convolutional networks," in *Proc. IEEE Int. Conf. Comput. Vis. (ICCV)*, Oct. 2017, pp. 764–773.
[41] X. Zhu, H. Hu, S. Lin, and J. Dai, "Deformable ConvNets v2: More deformable, better results," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2019, pp. 9308–9316.
[42] Y. Chen, H. Jiang, C. Li, X. Jia, and P. Ghamisi, "Deep feature extraction and classification of hyperspectral images based on convolutional neural networks," *IEEE Trans. Geosci. Remote Sens.*, vol. 54, no. 10, pp. 6232–6251, Oct. 2016.
[43] C. Shi and L. Wang, "Incorporating spatial information in spectral unmixing: A review," *Remote Sens. Environ.*, vol. 149, pp. 70–87, Jun. 2014.
[44] S. K. Roy, G. Krishna, S. R. Dubey, and B. B. Chaudhuri, "HybridSN: Exploring 3D–2D CNN feature hierarchy for hyperspectral image classification," *IEEE Geosci. Remote Sens. Lett.*, vol. 17, no. 2, pp. 277–281, Feb. 2020.
[45] M. Zhao, M. Wang, J. Chen, and S. Rahardja, "Hyperspectral unmixing for additive nonlinear models with a 3-D-CNN autoencoder network," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5509415.
[46] A. Hassani, S. Walton, J. Li, S. Li, and H. Shi, "Neighborhood attention transformer," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2023, pp. 6185–6194.
[47] Z. Xia, X. Pan, S. Song, L. E. Li, and G. Huang, "Vision transformer with deformable attention," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2022, pp. 4794–4803.



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